Another Physics and Route E: First-Principles Separation Theorems, an Exact Discriminant–Killing Bridge, and a Reversible Research Ledger

Route F research note

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Abstract

This ledger asks which parts of the manuscript *Another physics* can be connected to Route E without confusing an algebraic resemblance with a physical derivation. The literal identification of local energy density with time, phase-dependent disappearance of invariant mass, fundamental negative energy, and indestructible entanglement is ruled out under ordinary local, positive-energy relativistic quantum theory. A conditional complex-scalar completion does support finite-charge Q-balls, but its internal phase is not itself an observable clock.

The strongest exact connection is new but elementary. On Route E's H3$^+$-conditional branch, $H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathfrak{sl}_2(\mathbb{C})$. For a binary quadratic, its discriminant simultaneously detects two distinct zeros, a regular-semisimple Möbius flow, a nonzero Killing norm, and a nonzero Route-E second-transvectant contact. Thus the two-center divisor and K_{tr} are not merely parallel observations: they are two readings of the same invariant. A Fubini–Study moment map can further label the two fixed points by opposite bounded charges without introducing negative Hamiltonian energy.

A candidate action-level bridge makes the Route-E messenger mixing depend on a Q-ball field, $\lambda(\Phi) = g\Phi/\Lambda$, giving $\zeta_{\mathrm{eff}} = g^2\Phi^2/\Lambda^2$ and hence the same weight-two phase law as Route E. This is explicitly a bridge axiom, not a promoted result: an exact Q-ball $U(1)$ makes every invariant observable stationary, while an observable phase modulation requires relative-phase breaking and therefore reopens charge leakage and soliton stability. All claims are classified, derived, and tied to a reproducible numerical ledger; no physics promotion is authorized by the present note.

Contents

1 Provenance, scope, and logical contract

The source under review is the 15-page file `Another physics.pdf`, supplied from the local Downloads directory, with PDF metadata title `Another physics_20230910.docx` and SHA-256

`7355cf2237513a6e943555ac2720871a5e81915fd770541baea9d87eccf2654e`.

The Route-E reference state for this ledger is repository commit

`e7ba020227853a3b731235b5f3a42083559e7ef2`.

The governing Route-E sources are `route_E/tex/route_e_first_principles.tex`, `route_E/tex/three_families_killing_contact.tex`, and the canonical dynamics registry `route_E/code_dyn/dyn_claim_registry.json`.

Four logical labels are used throughout.

Label	Meaning
theorem	Consequence of explicitly stated mathematical or physical assumptions, with a proof in this ledger or an identified Route-E proof.
conditional completion	A consistent model after a new action, field content, or boundary condition has been chosen. It does not derive that choice from the source manuscript.
bridge axiom	An identification between objects belonging to two different models. Similar degree, phase, or dimension is not a proof of the identification.
numerical check	Reproducible arithmetic within a declared approximation. It corroborates formulas but cannot promote a bridge axiom.

Promotion gate 1.1 (Non-promotion). *The machine card may be green only with the non-promoting status* `mechanical_separation_checks_pass_no_physics_promotion`. *Its Boolean field* `physics_promotion_allowed` *must remain* `false` *until an action-level embedding, stability analysis, and all affected Route-E phenomenology gates pass.*

2 Covariant dictionary and literal-theory boundary

The source concepts are useful only after observer, sign, and conservation laws are made explicit. We use metric signature $(-+++)$. For an observer with unit timelike four-velocity u^μ, the local measured energy density is

$$E_d(x;u) := T_{\mu\nu}(x)u^\mu u^\nu. \tag{2.1}$$

This is a scalar after u is specified, but it is not an observer-independent replacement for spacetime or proper time. A viable information variable is instead a Noether charge, an entropy/correlation functional, or an order parameter. In the Q-ball completion below it is

$$E_i \quad \rightsquigarrow \quad Q_I = \int_\Sigma \mathrm{d}\Sigma_\mu \, j_I^\mu, \qquad \nabla_\mu j_I^\mu = 0. \tag{2.2}$$

The symbols E_p, E_n will later be reinterpreted as bounded moment-map polarities $\mu = +1, -1$, not as positive and negative Hamiltonian energies.

2.1 Local density cannot determine clock rate

Theorem 2.1 (Local-density time no-go). *No function of the pointwise scalar $E_d(x;u)$ alone can reproduce the gravitational clock rate in all solutions of general relativity.*

Proof. Consider a static thin spherical shell of mass M and radius $R > 2GM/c^2$. The vacuum interior is flat but its lapse is fixed by matching to the exterior:

$$\mathrm{d}s^2_{r<R} = -\left(1 - \frac{2GM}{Rc^2}\right)c^2\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\mathrm{d}\Omega^2. \tag{2.3}$$

Both the shell interior and spatial infinity have $T_{\mu\nu} = 0$, hence $E_d = 0$, but stationary clocks obey

$$\frac{\mathrm{d}\tau_{\mathrm{in}}}{\mathrm{d}t_\infty} = \sqrt{1 - \frac{2GM}{Rc^2}} \neq 1. \tag{2.4}$$

Two points with the same local density therefore have different clock rates. The missing datum is the nonlocal metric solution/boundary condition, not a different definition of local energy density. □

For Earth-scale numbers, $GM/(Rc^2) = 6.9613 \times 10^{-10}$, corresponding at leading order to about $60.15\,\mu$s per day. In the laboratory weak-field limit,

$$\frac{\Delta\nu}{\nu} = \frac{g\Delta h}{c^2} = 1.091 \times 10^{-18}\left(\frac{\Delta h}{1\,\mathrm{cm}}\right). \tag{2.5}$$

This scale has been directly tested by a centimeter atomic-clock network [1] and is an immediate constraint on any replacement for metric proper time.

The same local E_d can also curve spacetime differently because pressure matters. Tracing

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \tag{2.6}$$

and using $T = -\varepsilon + 3p$ gives

$$R = 4\Lambda + \frac{8\pi G}{c^4}(\varepsilon - 3p). \tag{2.7}$$

Dust and radiation may have the same ε, but $p = 0$ and $p = \varepsilon/3$ give different Ricci scalars. A density-only clock or gravity law therefore discards essential stress data.

2.2 Mass, energy sign, and entanglement

Theorem 2.2 (Phase-toggle no-go in a fixed particle representation)**.** *For a stable one-particle irreducible representation of the Poincaré group, the rest mass cannot alternate between $m \neq 0$ and $m = 0$ as an unobserved internal phase changes while the state remains in that representation.*

Proof. The mass is fixed by the Casimir

$$P_\mu P^\mu = -m^2c^2. \tag{2.8}$$

It commutes with the representation. If a phase trajectory changed the eigenvalue from $-m^2c^2$ to zero, it would leave the massive irreducible representation and enter a massless one. A periodic *coupling*, spectral weight, quasiparticle gap, or detector response may vanish; invariant rest mass in a fixed representation may not. □

Likewise, a literal independent negative-energy oscillator,

$$H = \omega a^\dagger a - \Omega b^\dagger b, \tag{2.9}$$

has $\inf \operatorname{spec} H = -\infty$, since $\langle n_b|H|n_b\rangle = -n_b\Omega$. A stable completion must reinterpret the sign as charge, orientation, a hole relative to a filled state, or an effective-medium variable; it cannot be a freely excitable fundamental negative-energy sector.

Entanglement is not indestructible either. For

$$\rho_p = (1-p)|\Phi^+\rangle\langle\Phi^+| + \frac{p}{4}I_4, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \tag{2.10}$$

the partial transpose eigenvalues are

$$\operatorname{spec}(\rho_p^{T_B}) = \left\{\frac{2-p}{4}, \frac{2-p}{4}, \frac{2-p}{4}, \frac{-2+3p}{4}\right\}. \tag{2.11}$$

For $p \geq 2/3$ the state is PPT and, in 2×2, separable. What is forbidden is superluminal signalling, not dynamical loss of entanglement.

2.3 Other immediate separation results

For a massive electron of finite Lorentz factor γ,

$$\beta := \frac{v}{c} = \sqrt{1-\gamma^{-2}} < 1. \tag{2.12}$$

Magnetism is encoded in the gauge curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, not in electrons moving exactly at c. Event-horizon escape is similarly excluded by causal structure; jets must be launched outside the horizon.

If a proposed galactic potential is parabolic, $\Phi_N(r) = ar^2$, then

$$\frac{v_c^2}{r} = \frac{\mathrm{d}\Phi_N}{\mathrm{d}r} = 2ar \quad \Longrightarrow \quad v_c = \sqrt{2a}\, r, \tag{2.13}$$

which is solid-body rotation, not an asymptotically flat curve. A concrete modified-gravity alternative must fit both dynamics and lensing; recent weak lensing reaches far beyond classical rotation-curve radii [2]. Finally a decay half-life alone supplies no force. Only momentum flux does; for perfectly absorbed radiation

$$F = \frac{P}{c} = 3.3356\,\mathrm{nN}\left(\frac{P}{1\,\mathrm{W}}\right). \tag{2.14}$$

3 The Route-E theorem boundary imported into Route F

Route E assumes a self-carried family space

$$V = H^0(\Sigma, L), \qquad L \simeq T_\Sigma, \qquad V \simeq H^0(\Sigma, T_\Sigma) = \mathfrak{aut}(\Sigma). \tag{3.1}$$

The original invariant-contact hypothesis H3 requires a nondegenerate symmetric invariant bilinear form. It gives only $N_{\mathrm{fam}} \leq 3$, since the genus-one one-dimensional Abelian algebra admits such a form. The exact selection $N_{\mathrm{fam}} = 3$ requires the explicit stronger hypothesis H3$^+$: the contact is the nondegenerate adjoint-trace/Killing form. On that selected branch,

$$\Sigma \simeq \mathbb{CP}^1, \qquad T_{\mathbb{CP}^1} \simeq \mathcal{O}(2), \qquad V \simeq \mathfrak{sl}_2(\mathbb{C}), \qquad \dim_{\mathbb{C}} V = 3. \tag{3.2}$$

Thus every result below that invokes this branch is conditional on H3$^+$; the subsequent algebra needs no additional physical axiom once the branch is chosen.

Riemann–Roch and Serre duality give the dimension transparently:

$$h^0(\mathcal{O}(2)) - h^1(\mathcal{O}(2)) = \deg \mathcal{O}(2) + 1 = 3, \tag{3.3}$$

$$h^1(\mathcal{O}(2)) = h^0(K \otimes \mathcal{O}(-2)) = h^0(\mathcal{O}(-4)) = 0, \tag{3.4}$$

so $h^0 = 3$. Here $\mathcal{O}(2)$ is a degree-two holomorphic line bundle, not the orthogonal group $O(2)$.

In Route E's spherical basis (T_{+1}, T_0, T_{-1}),

$$K_{\mathrm{tr}} = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \qquad B = 2\sqrt{3}\, K_{\mathrm{tr}}, \qquad K_{\mathrm{tr}}^2 = \frac{1}{3} I. \tag{3.5}$$

This complex symmetric invariant pairing is not a Hermitian energy operator. Its indefinite real-slice signs cannot be read as positive and negative physical energies.

4 Exact discriminant–two-center–Killing theorem

4.1 Global vector fields and the adjoint representation

Let $\xi = v/u$ on an affine chart of $\mathbb{CP}^1$. A holomorphic vector field has the local form $X = q(\xi)\partial_\xi$. On the other chart $\eta = 1/\xi$,

$$q(\xi)\partial_\xi = -\eta^2 q(1/\eta)\partial_\eta. \tag{4.1}$$

Regularity at $\eta = 0$ therefore forces $\deg q \le 2$:

$$X_q = (a + b\xi + c\xi^2)\partial_\xi. \tag{4.2}$$

The basis

$$E = \partial_\xi, \qquad H = -2\xi\partial_\xi, \qquad F = -\xi^2\partial_\xi \tag{4.3}$$

obeys

$$[H, E] = 2E, \qquad [H, F] = -2F, \qquad [E, F] = H, \tag{4.4}$$

which proves $H^0(\mathbb{CP}^1, T) \simeq \mathfrak{sl}_2(\mathbb{C})$.

Homogenizing (4.2) gives the binary quadratic

$$p(u, v) = au^2 + buv + cv^2, \qquad \Delta(p) = b^2 - 4ac. \tag{4.5}$$

The representation statement is precisely

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathrm{Sym}^2(\mathbb{C}^2)^*, \tag{4.6}$$

and the invariant epsilon tensor identifies the dual with $\mathrm{Sym}^2\,\mathbb{C}^2$ for $SL(2)$.

Theorem 4.1 (Discriminant–Killing equivalence)**.** *For a nonzero $X_q \in H^0(\mathbb{CP}^1, T)$, the following are equivalent:*

1. *its zero divisor is a sum of two distinct points;*
2. *$\Delta(p) \neq 0$;*
3. *the corresponding $\mathfrak{sl}_2(\mathbb{C})$ element is regular semisimple;*
4. *its Killing norm is nonzero;*
5. *its Route-E contact norm $K_{\mathrm{tr}}(X_q, X_q)$ is nonzero.*

The complement $\Delta = 0$ is the nonzero nilpotent/contact-null cone and has one double zero.

Proof. Take the standard projective infinitesimal matrix

$$A_q = \begin{pmatrix} -b/2 & -c \\ a & b/2 \end{pmatrix}. \tag{4.7}$$

Direct multiplication gives

$$A_q^2 = \frac{\Delta(p)}{4} I. \tag{4.8}$$

Hence $\Delta \neq 0$ gives two nonzero opposite eigenvalues and a regular semisimple element. If $\Delta = 0$ and $A_q \neq 0$, then $A_q^2 = 0$, so the element is nilpotent. The roots of p are distinct exactly when its discriminant is nonzero; when it vanishes, the homogeneous quadratic is a square and has a double projective root. Counting multiplicity, the total degree is always

$$\deg \operatorname{div}(X_q) = c_1(T_{\mathbb{CP}^1})[\mathbb{CP}^1] = \chi(S^2) = 2. \tag{4.9}$$

For $\mathfrak{sl}_2(\mathbb{C})$, the canonical adjoint-trace Killing form obeys $B(A, A) = 4\,\mathrm{Tr}_{2\times 2}(A^2)$, so the displayed matrix gives

$$B(A_q, A_q) = 2\Delta(p). \tag{4.10}$$

No rescaling of Lie-algebra generators is allowed here. Instead, write the same matrix in Route E's spherical basis as

$$A_q = x_+T_{+1} + x_0T_0 + x_-T_{-1} = \begin{pmatrix} x_0/2 & -x_+/\sqrt{2} \\ x_-/\sqrt{2} & -x_0/2 \end{pmatrix}, \qquad a = \frac{x_-}{\sqrt{2}}, \quad b = -x_0, \quad c = \frac{x_+}{\sqrt{2}}. \tag{4.11}$$

Then $\Delta = x_0^2 - 2x_+x_-$, while Eq. (3.5) gives $B(\boldsymbol{x}, \boldsymbol{x}) = 2\sqrt{3}\,\boldsymbol{x}^T K_{\mathrm{tr}}\boldsymbol{x} = 2(x_0^2 - 2x_+x_-) = 2\Delta$. This is an invertible change of coordinates, not a change of bracket. It proves the remaining zero-versus-nonzero equivalence and fixes the normalization. □

The normalization can be checked without words. Write

$$p = \frac{x_-}{\sqrt{2}}u^2 - x_0uv + \frac{x_+}{\sqrt{2}}v^2, \qquad \boldsymbol{x} = (x_+, x_0, x_-)^T. \tag{4.12}$$

Then

$$\Delta(p) = x_0^2 - 2x_+x_-, \tag{4.13}$$

$$\boldsymbol{x}^T K_{\mathrm{tr}}\boldsymbol{x} = \frac{1}{\sqrt{3}}(x_0^2 - 2x_+x_-), \tag{4.14}$$

$$\boxed{B(\boldsymbol{x},\boldsymbol{x}) = 2\Delta(p) = 2\sqrt{3}\,\boldsymbol{x}^T K_{\mathrm{tr}}\boldsymbol{x}.} \tag{4.15}$$

The unnormalized second transvectant is

$$(p,p)_2 = p_{uu}p_{vv} - 2p_{uv}^2 + p_{vv}p_{uu} = 8ac - 2b^2 = -2\Delta(p) = -B(\boldsymbol{x},\boldsymbol{x}), \tag{4.16}$$

so the same invariant is the transvectant singlet, again up to its declared normalization.

Remark 4.2 (What the theorem does not say)**.** *The theorem welds two Route-E structures exactly. It does not identify the two zeros with positive/negative energy, two particles, or two families. It also shows why the regular-semisimple qualifier matters: a generic nonzero section has two distinct centers, but a nilpotent section has one center with multiplicity two.*

5 A safe two-polarity bridge: Hopf reduction and moment map

A creative way to retain the source manuscript's two polarities without a negative-energy instability is to begin with a complex doublet $z = (z_0, z_1) \in \mathbb{C}^2$. At fixed norm, $z^\dagger z = 1$, the state space is S^3. Quotienting the common phase gives

$$S^3/U(1) \simeq \mathbb{CP}^1. \tag{5.1}$$

The Hopf map is

$$\boldsymbol{n} = \left(2\Re z_0^* z_1,\ 2\Im z_0^* z_1,\ |z_0|^2 - |z_1|^2\right), \qquad \boldsymbol{n}^2 = 1. \tag{5.2}$$

For the relative circle action, a normalized Fubini–Study moment map is

$$\mu([z_0 : z_1]) = \frac{|z_0|^2 - |z_1|^2}{|z_0|^2 + |z_1|^2}, \qquad -1 \le \mu \le 1. \tag{5.3}$$

Its two fixed points satisfy

$$p_+ = [1:0], \quad \mu(p_+) = +1; \qquad p_- = [0:1], \quad \mu(p_-) = -1. \tag{5.4}$$

The numerical endpoint values ± 1 use the displayed generator and the symmetric additive normalization. Rescaling the generator by c rescales them to $\pm c$. What is normalization-independent is that the compact moment image is bounded and that the two fixed points are opposite extrema (up to reversing orientation). This suggests the normalized safe dictionary

$$E_p, E_n \quad \rightsquigarrow \quad \mu = +1, -1 \quad \text{(charge/orientation polarity)}, \tag{5.5}$$

while the physical Hamiltonian remains bounded below.

With

$$\omega_{\mathrm{FS}} = \frac{\mathrm{i}\,\mathrm{d}\xi \wedge \mathrm{d}\bar{\xi}}{(1+|\xi|^2)^2}, \qquad \frac{1}{2\pi}\int_{\mathbb{CP}^1} \omega_{\mathrm{FS}} = 1, \tag{5.6}$$

geometric quantization of Chern number $k \geq 0$ gives

$$L \simeq \mathcal{O}(k), \qquad \mathcal{H}_k = H^0(\mathbb{CP}^1, \mathcal{O}(k)) \simeq \mathrm{Sym}^k\,\mathbb{C}^2, \qquad \dim \mathcal{H}_k = k+1. \tag{5.7}$$

If, and only if, the physical bubble reduction also satisfies Route E's self-carried identification $L = T_{\mathbb{CP}^1} \simeq \mathcal{O}(2)$, then $k = 2$ and $\dim \mathcal{H} = 3$.

Bridge axiom 5.1 (Projective-doublet bridge, BA-P)**.** *The energy-information order parameter is a physical complex doublet; its fixed-charge reduction is $\mathbb{CP}^1$; its Berry/prequantum bundle equals $T_{\mathbb{CP}^1}$; and its chiral zero-mode operator has $\ker D^+ \simeq H^0(T)$, $\ker D^- = 0$, with no extra charged zero modes.*

None of the four clauses follows merely from the Hopf identity (5.1). This bridge is therefore a concrete research target, not a derivation of three families.

6 Conditional Q-ball completion of an energy bubble

6.1 Action, charge, and radial equation

Replace a literal positive/negative-energy bubble by one complex scalar with a healthy kinetic term:

$$S_\Phi = \int \mathrm{d}^4x\sqrt{-g}\left[-g^{\mu\nu}\partial_\mu\Phi^*\partial_\nu\Phi - U(|\Phi|)\right], \tag{6.1}$$

with

$$U(f) = m^2 f^2 - \lambda f^4 + \frac{\eta}{M^2} f^6, \qquad m^2, \lambda, \eta, M^2 > 0. \tag{6.2}$$

The global $U(1)$ current can be normalized so that for

$$\Phi(t,r) = f(r)e^{\mathrm{i}\omega t} \tag{6.3}$$

the charge and energy in flat space are

$$Q = 2\omega \int \mathrm{d}^3x\, f^2, \tag{6.4}$$

$$E = \int \mathrm{d}^3x \left[\omega^2 f^2 + (\boldsymbol{\nabla} f)^2 + U(f)\right]. \tag{6.5}$$

Variation gives the boundary-value problem

$$f'' + \frac{2}{r}f' = \left(m^2 - \omega^2 - 2\lambda f^2 + \frac{3\eta}{M^2}f^4\right) f, \qquad f'(0) = 0, \quad f(\infty) = 0. \tag{6.6}$$

Proposition 6.1 (Existence window for the sextic model)**.** *The thin-wall Q-ball window is nonempty when*

$$0 < \omega_0^2 := \min_{f>0} \frac{U(f)}{f^2} < m^2, \tag{6.7}$$

and for (6.2)

$$f_0^2 = \frac{\lambda M^2}{2\eta}, \qquad \omega_0^2 = m^2 - \frac{\lambda^2 M^2}{4\eta}, \qquad \omega_0^2 < \omega^2 < m^2. \tag{6.8}$$

Proof. The effective ratio is

$$\frac{U(f)}{f^2} = m^2 - \lambda f^2 + \frac{\eta}{M^2} f^4. \tag{6.9}$$

Differentiation with respect to f^2 gives $-\lambda + 2\eta f^2/M^2 = 0$, hence the displayed f_0 and ω_0. The upper inequality ensures exponential decay at infinity because the linearized equation has inverse length $\sqrt{m^2 - \omega^2}$; the lower inequality makes the effective potential $U_\omega = U - \omega^2 f^2$ negative somewhere, allowing a nontrivial bounce-like radial trajectory. This is the standard Q-ball criterion [3]; interacting extensions can alter the stability range and must be retested [4]. □

6.2 Reproducible benchmark and approximation boundary

For

$$m = M = 1\,\mathrm{GeV}, \qquad \lambda = \eta = 1, \qquad Q = 10^6, \tag{6.10}$$

one finds

$$f_0 = \frac{1}{\sqrt{2}}\,\mathrm{GeV}, \qquad \omega_0 = \frac{\sqrt{3}}{2}\,\mathrm{GeV} = 0.8660254\,\mathrm{GeV}. \tag{6.11}$$

At $\omega = \omega_0$,

$$U_\omega(f) = f^2(f^2 - f_0^2)^2, \tag{6.12}$$

and the wall tension in the present kinetic convention is

$$\sigma = 2\int_0^{f_0} \sqrt{U_\omega(f)}\,\mathrm{d}f = \frac{1}{8}\,\mathrm{GeV}^3. \tag{6.13}$$

Using $Q = 2\omega_0 f_0^2(4\pi R^3/3)$ gives

$$R_0 = \left(\frac{3Q}{8\pi\omega_0 f_0^2}\right)^{1/3} = 65.0819045\,\mathrm{GeV}^{-1} = 12.8424157\,\mathrm{fm}. \tag{6.14}$$

Adding the leading surface energy yields

$$\frac{E}{Q} \simeq \omega_0 + \frac{4\pi R_0^2\sigma}{Q} = 0.872678754\,\mathrm{GeV} < m. \tag{6.15}$$

An independent fixed-step-profile variation minimizes

$$E(R) = \frac{Q^2}{4f_0^2 V(R)} + U(f_0)V(R) + 4\pi R^2\sigma, \qquad V(R) = \frac{4\pi R^3}{3}, \tag{6.16}$$

and gives

$$R = 64.9712654\,\mathrm{GeV}^{-1}, \quad \omega = 0.870457184\,\mathrm{GeV}, \quad E/Q = 0.872667434\,\mathrm{GeV}. \tag{6.17}$$

These are thin-wall/step-profile checks, not an exact numerical solution of (6.6). The exact profile and fluctuation operator remain a promotion gate.

6.3 Exact-symmetry phase no-observable theorem

Theorem 6.2 (No phase clock from an exact Q-ball symmetry)**.** *Let the action and a local observable $\mathcal{O}$ be invariant under the global transformation $\Phi \mapsto e^{\mathrm{i}\alpha}\Phi$. On the stationary Q-ball orbit $\Phi = f(r)e^{\mathrm{i}\omega t}$, $\mathcal{O}$ is independent of time. Therefore the exact Q-ball phase alone cannot make an invariant mass or visibility periodically appear and disappear.*

Proof. For every t, the field is obtained from $f(r)$ by the symmetry transformation with $\alpha = \omega t$. Thus

$$\mathcal{O}[f(r)e^{\mathrm{i}\omega t}] = \mathcal{O}[e^{\mathrm{i}\omega t}f(r)] = \mathcal{O}[f(r)]. \tag{6.18}$$

In particular, $|\Phi|^2$, $T_{\mu\nu}$, and any invariant pole mass are stationary. An observable phase requires a second phase reference or an explicitly symmetry-breaking spurion. □

This creates a useful no-free-lunch constraint. Exact $U(1)$ protects Q but hides the absolute phase; a fixed relative-phase term can expose the phase but then generally gives $\nabla_\mu j^\mu \neq 0$, so absolute Q-ball stability is replaced by a calculable lifetime.

7 Route-E weight-two phase and a physical relative phase

Route E reconstructs

$$M_R = -m_D^T m_\nu^{-1} m_D, \qquad m_D = vY_\nu. \tag{7.1}$$

Under a uniform nonzero complex rescaling $Y_\nu \mapsto yY_\nu$,

$$M_R \mapsto y^2 M_R. \tag{7.2}$$

If the positive scale is defined by a Takagi singular value, $M_* = \sigma_{\max}(M_R) > 0$, then

$$M_* \mapsto |y|^2 M_*. \tag{7.3}$$

For the normalized contact projection this gives the exact covariance

$$\boxed{\zeta \mapsto \frac{y^2}{|y|^2}\zeta = e^{2\mathrm{i}\arg y}\zeta.} \tag{7.4}$$

Consequently ζ is invariant under any nonzero real uniform scaling, not only positive scaling; for a complex scaling its phase has weight two. The normalized construction is undefined at $y = 0$, and family-dependent rescalings do not obey (7.4).

The benchmark is

$$\zeta = 0.1076472949 + 0.0736514853\mathrm{i}, \qquad |\zeta| = 0.1304319033, \qquad \arg\zeta = 0.6000380203. \tag{7.5}$$

Uniform phase covariance is not itself observable. If Φ carries weight one and ζ weight two, a rephasing-invariant relative phase is

$$e^{\mathrm{i}\delta} := \frac{\zeta^*\Phi^2}{|\zeta|\,|\Phi|^2}, \qquad \delta = 2\arg\Phi - \arg\zeta. \tag{7.6}$$

A bounded visibility ansatz with the source manuscript's four special phases is then

$$w(\delta) = \frac{1+\cos\delta}{2} = \cos^2\left(\arg\Phi - \frac{1}{2}\arg\zeta\right), \qquad 0 \le w \le 1. \tag{7.7}$$

This changes a coupling or spectral residue, not the Poincaré Casimir. The two naive identifications are numerically inequivalent:

$$\cos^2(\arg\zeta) = 0.68114344, \qquad \cos^2(\arg\zeta/2) = 0.91265707. \tag{7.8}$$

Route E selects neither convention.

For two complex symmetric Majorana structures $M_R = M_V + M_C$, $M_C = \zeta K_{\mathrm{tr}}$, a family-basis congruence sends both as $M \mapsto U^T M U$. If M_V is invertible,

$$\mathcal{R} := M_V^{-1} M_C \mapsto U^{-1}\mathcal{R}U. \tag{7.9}$$

Therefore quantities such as

$$\Im\,\mathrm{Tr}(\mathcal{R}^n), \qquad \arg\det(I + \mathcal{R}) \tag{7.10}$$

are basis invariants that can test a *relative* contact phase. Any future phase claim must be written in such invariant form rather than as an absolute $\arg\zeta$.

8 Candidate Q-ball–Schur-complement bridge

8.1 Action-level construction

Let $N \in V$ and a sterile heavy messenger $X \in V^*$ be left-handed Weyl family vectors. A non-supersymmetric quadratic mass sector with a dynamical scalar-dependent mixing is

$$\begin{aligned}\mathcal{L}_{NX} &= \mathrm{i}N^\dagger\bar{\sigma}^\mu D_\mu N + \mathrm{i}X^\dagger\bar{\sigma}^\mu\partial_\mu X \\ &\quad - \frac{M_*}{2}\left[N^T M_V N + 2\lambda(\Phi)X^T N - X^T K_{\mathrm{tr}}^{-1} X\right] + \mathrm{h.c.}\end{aligned} \tag{8.1}$$

At momenta well below the messenger gap, the algebraic equation is

$$X = \lambda(\Phi) K_{\mathrm{tr}} N. \tag{8.2}$$

Substitution gives

$$\mathcal{L}_{\mathrm{eff}} \supset -\frac{M_*}{2} N^T\left[M_V + \lambda(\Phi)^2 K_{\mathrm{tr}}\right]N + \mathrm{h.c.} \tag{8.3}$$

More generally, if the messenger kernel is $-\mu K_{\mathrm{tr}}^{-1}$, then

$$\zeta_{\mathrm{eff}} = \frac{\lambda(\Phi)^2}{\mu}. \tag{8.4}$$

Route E sets $\mu = 1$ by a field rescaling; keeping it visible is useful for charge and normalization audits.

The direct phase-doubling bridge is

$$\lambda(\Phi) = g\frac{\Phi}{\Lambda} \quad\Longrightarrow\quad \boxed{\zeta_{\mathrm{eff}}(x) = g^2\frac{\Phi(x)^2}{\Lambda^2}.} \tag{8.5}$$

On a Q-ball background,

$$\zeta_{\mathrm{eff}}(r,t) = g^2\frac{f(r)^2}{\Lambda^2}e^{2\mathrm{i}\omega t}. \tag{8.6}$$

This exactly reproduces a localized weight-two coefficient. It is the most direct candidate relation between the two research programs.

Bridge axiom 8.1 (Q-ball contact portal, BA-QK)**.** *The physical heavy mode has the Route-E family representation and kernel in* (8.1)*; the dynamical mixing is $g\Phi/\Lambda$; all charge, gauge, anomaly, and operator-selection rules are satisfied; and the resulting relative phase cannot be removed by a field redefinition.*

8.2 Why the portal is not yet a completion

The following gates are mandatory.

Charge ledger. With $q_\Phi = +1$, the choices $q_X = 0$, $q_N = -1$ make ΦXN and XX invariant, but $N^T M_V N$ then needs a charge-$+2$ spurion. If M_V is instead a fixed explicit term, the Q-ball $U(1)$ is broken. If all spurions transform dynamically, the full background may be stationary in a rotating frame and no visibility modulation follows. This tension must be resolved, not hidden in notation.

Tree-level normalization. For canonical messenger kinetic terms, the leading low-energy substitution also gives

$$Z_N = I + |\lambda|^2 K_{\mathrm{tr}}^\dagger K_{\mathrm{tr}} = \left(1 + \frac{|\lambda|^2}{3}\right) I. \tag{8.7}$$

At the Route-E benchmark $|\lambda|^2 = |\zeta|$,

$$\delta Z_N = \frac{|\zeta|}{3} = 0.0434773011. \tag{8.8}$$

This is a tree-level matching effect. The canonical physical test is the complete Weinberg combination $YM^{-1}Y^T$, not ζ by itself.

Full propagator and interaction. When external energies are not negligible relative to all heavy Takagi singular values, one must use

$$\mathcal{M}_{NX} = M_* \begin{pmatrix} M_V & \lambda I \\ \lambda I & -K_{\mathrm{tr}}^{-1} \end{pmatrix} \tag{8.9}$$

and its kinetic matrix. The original fixed-λ model is Gaussian and does not by itself generate a $|\lambda|^2/(16\pi^2)$ anomalous dimension. The dynamical vertex $g\Phi XN/\Lambda$ supplies a genuine interaction, but its loop matching, cutoff power counting, and counterterms must be calculated.

Selection and leakage. The existing Route-B rule permits the unwanted post-breaking operator XLH. A valid discrete/gauge rule must forbid it and enumerate operators through a declared dimension. Moreover, if neutrinos or messengers carry the Q-ball charge, the soliton may emit them. Absolute stability requires its energy per unit charge to lie below every kinematically available charge-carrying channel; explicit phase locking instead requires a lifetime calculation.

Static benchmark versus rotating background. Route E fits a constant ζ. Equation (8.6) is periodic relative to a fixed M_V. It therefore requires a Floquet/adiabatic seesaw analysis and cannot be inserted into the static fit without rerunning neutrino, threshold, proton-decay, and cosmology gates.

9 Candidate dynamical origin of H3$^+$

The previous sections use H3$^+$ but do not derive it. There is, however, a specific loop mechanism worth testing. Couple a complete heavy family adjoint multiplet to a background family source A^a_μ. Covariance and the Ward identity constrain its vacuum polarization to

$$\Pi^{\mu\nu}_{ab}(p) = (p^2\eta^{\mu\nu} - p^\mu p^\nu)\,\Pi(p^2)\ \mathrm{Tr}_{\mathrm{ad}}(\mathrm{ad}\,T_a\,\mathrm{ad}\,T_b). \tag{9.1}$$

The family tensor is therefore

$$B_{ab}^{\mathrm{Kill}} := \mathrm{Tr}_{\mathrm{ad}}(\mathrm{ad}\, T_a\, \mathrm{ad}\, T_b). \tag{9.2}$$

For a simple algebra, invariant symmetric bilinears form a one-dimensional space, so a nonzero result is proportional to the Killing form. On the Route-E branch this is $2\sqrt{3}K_{\mathrm{tr}}$.

Bridge axiom 9.1 (Adjoint-loop H3$^+$ mechanism, BA-H3)**.** *A healthy UV sector contains a complete, non-canceling adjoint current whose low-energy two-point function induces the required family bilinear, and a specified source/messenger map converts that bilinear into the Majorana contact rather than merely a gauge kinetic term.*

Equation (9.1) explains why a Killing tensor is natural once an adjoint loop exists; it does not prove the existence of that loop, its nonzero coefficient, a positive spectral density in the relevant channel, or the conversion to a Majorana mass. H3$^+$ therefore remains physically open.

10 The oscillatory formula: exact solution and strict separation

The source manuscript includes

$$\sin(x^2) + \cos(x^2) = \sin(y^2) + \cos(y^2). \tag{10.1}$$

Using $\sin t + \cos t = \sqrt{2}\sin(t + \pi/4)$, the identity $\sin A = \sin B$ gives

$$A = B + 2\pi n \quad \text{or} \quad A = \pi - B + 2\pi n, \qquad n \in \mathbb{Z}. \tag{10.2}$$

With $A = x^2 + \pi/4$, $B = y^2 + \pi/4$, the complete real solution set is

$$\boxed{x^2 - y^2 = 2\pi n} \qquad \text{or} \qquad \boxed{x^2 + y^2 = \frac{\pi}{2} + 2\pi n}, \tag{10.3}$$

subject to real-coordinate nonemptiness. The circular radii are

$$r_n^2 = \frac{\pi}{2} + 2\pi n, \qquad n \geq 0, \tag{10.4}$$

and their exact radial pitch is

$$r_{n+1} - r_n = \frac{2\pi}{r_{n+1} + r_n} \sim \frac{\pi}{r_n}. \tag{10.5}$$

Thus a shrinking interference pitch is a theorem. These are continuous circles/hyperbolae, not discrete atomic orbitals; standard subshell degeneracies are $2(2\ell + 1) = 2, 6, 10, 14, \ldots$.

There is no direct Route-E identification. The functions $\sin(\xi^2)$ and $\cos(\xi^2)$ have an essential singularity in the $\eta = 1/\xi$ chart, whereas global sections of $T_{\mathbb{CP}^1}$ are exactly degree-at-most-two polynomials. A shared appearance of squares is insufficient to identify the state spaces.

11 Relational time and gravity: parallel interfaces, not Route-E derivations

The source intuition that time may be relational has a standard conditional form. Let a clock C and system S satisfy the stationary constraint

$$(H_C + H_S)|\Psi\rangle = 0. \tag{11.1}$$

If clock states obey $\langle t|H_C = -\mathrm{i}\hbar\partial_t\langle t|$, then

$$|\psi(t)\rangle := \langle t|\Psi\rangle \tag{11.2}$$

satisfies

$$\begin{aligned} 0 &= \langle t|(H_C + H_S)|\Psi\rangle && (11.3)\\ &= -\mathrm{i}\hbar\partial_t|\psi(t)\rangle + H_S|\psi(t)\rangle, && (11.4)\end{aligned}$$

hence

$$\mathrm{i}\hbar\partial_t|\psi(t)\rangle = H_S|\psi(t)\rangle. \tag{11.5}$$

This Page–Wootters construction [5] has active modern extensions, including finite-dimensional time-dilation mechanisms [6]. It does not identify time with local E_d, nor does it derive Route E.

A thermodynamic route to Einstein's equation is likewise conditional. On every local Rindler horizon, impose

$$\delta Q = T_U\delta S, \qquad T_U = \frac{\hbar\kappa}{2\pi}, \qquad \delta S = \frac{\delta A}{4G\hbar}. \tag{11.6}$$

The Raychaudhuri equation relates the area variation to $R_{\mu\nu}k^\mu k^\nu$, while the heat flux is proportional to $T_{\mu\nu}k^\mu k^\nu$. Requiring the equality for every null vector k^μ gives

$$R_{\mu\nu} + \Phi g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{11.7}$$

Stress conservation and the Bianchi identity fix $\Phi = -R/2 + \Lambda$, yielding Einstein's equation [7, 8]. The horizon entropy law and local equilibrium are inputs. This route is a gravity interface for later work, not evidence for the family/contact bridge.

12 Claim ledger and ordered research program

Claim	Status	Boundary / next gate
E_d alone determines time/gravity	refuted	Thin-shell and pressure counterexamples, Eqs. (2.3)–(2.7).
Invariant mass disappears at four internal phases	refuted	Poincaré Casimir; only coupling/residue/gap may be modulated.
Route-E CP1/O(2) carrier has three sections	conditional theorem	Requires H3$^+$ to select the branch; H3 alone gives only the upper bound.
Two centers $\Leftrightarrow \Delta \neq 0 \Leftrightarrow K_{\mathrm{tr}}$-norm nonzero	proved	Exact on the selected CP1 branch, Theorem 4.1.
Opposite center labels without negative energy	candidate bridge	Moment-map polarity is exact; physical identification requires BA-P.
Energy bubble as a Q-ball	conditional completion	Exact radial solution, Hessian, charge leakage, and gravity remain open.
ζ carries phase weight two	proved in Route E	Uniform complex rescaling only; the absolute phase is not observable.
$\zeta_{\mathrm{eff}} = g^2\Phi^2/\Lambda^2$	candidate bridge	Requires BA-QK, charge/selection rules, full matching, and rerun fits.

Claim	Status	Boundary / next gate
Adjoint loop physically derives H3$^+$	open mechanism	Killing tensor follows conditionally; existence and Majorana conversion do not.
Oscillatory equation quantizes atoms or gives CP1 sections	refuted	It gives continuous conics and has an essential singularity at CP1 infinity.
Relational time / thermodynamic gravity	parallel conditional tracks	Useful interfaces; neither derives Route E or the bubble map.

The sequential program is deliberately fail-closed.

AP–E1. Projective-doublet action. Write a two-component bubble action, identify its constraint and gauge/global quotient, and prove or refute that the physical reduced moduli space is $\mathbb{CP}^1$.

AP–E2. Discriminant/contact verifier. Preserve Theorem 4.1 as an exact symbolic regression test, including the regular/nilpotent boundary and Route-E normalization.

AP–E3. Self-carried geometric quantization. Derive the Berry curvature and Chern number from AP–E1 rather than imposing them; test whether the prequantum bundle is actually $T_{\mathbb{CP}^1}$.

AP–E4. Chiral zero modes. Construct the fluctuation/Dirac operator, prove $\ker D^+ \simeq H^0(T)$, compute $\ker D^-$, extra modes, and the spectral gap.

AP–E5. Exact Q-ball solution. Solve Eq. (6.6), compute $Q(\omega)$, $E(Q)$, fixed-charge Hessian modes, gravitational compactness, and all charge-emission thresholds.

AP–E6. Messenger/contact realization. Build a gauge- and anomaly-consistent version of Eq. (8.1); enumerate operators, forbid XLH, and verify the full kinetic plus six-by-six matching.

AP–E7. Physical-phase quotient. Remove all field rephasings, list basis-invariant CP quantities such as Eq. (7.10), and test whether a relative phase survives without destroying the Q-ball lifetime.

AP–E8. H3$^+$ origin test. Compute the candidate adjoint current correlator and its spectral sign; show whether it yields a Majorana contact or only a kinetic bilinear.

AP–E9. Integrated phenomenology. Rerun Route-E flavor/seesaw, threshold, proton-decay, cosmology, and time-dependent/Floquet gates with a single action. Only this stage can request physics promotion.

AP–E10. Independent gravity track. Constrain any E_d carrier with clock redshift, fifth-force, rotation-curve, and lensing data; do not merge it with AP–E1–9 without an explicit operator connecting the sectors.

13 2026-07-16 AP–E3 global/nonlinear and AP–E4 SQM handoff

This section is a reversible index into three standalone derivation notes. It records the equations needed to reconstruct the decision boundary without replacing their complete proofs and numerical ledgers.

13.1 Charged two-colour nonlinear proxy

The declared six-real-field diagnostic potential is

$$V = \frac{\lambda}{4}\left(\sigma^2 + \boldsymbol{\pi}^2 + 2|\Delta|^2 - v^2\right)^2 - h\sigma + \mu_g^2|\Delta|^2, \qquad q_\Delta = 2. \tag{13.1}$$

At the mesonic stationary point, $h/f = m_\pi^2$, and its exact Hessian contains

$$m_\sigma^2 = m_\pi^2 + 2\lambda f^2, \qquad m_\Delta^2 = m_\pi^2 + \mu_g^2. \tag{13.2}$$

The requested $(e_g, v_g/\Lambda_c)$ scan is explicitly a matching ansatz, not a UV loop result:

$$\frac{\mu_g^2}{\Lambda_c^2} = \frac{\mu_{\mathrm{PG},0}^2}{\Lambda_c^2} + c_g\left(e_g\frac{v_g}{\Lambda_c}\right)^2. \tag{13.3}$$

The mesonic nonlinear sector solves the $B = 1$ hedgehog with

$$B = -\frac{2}{\pi}\int_0^\infty \sin^2 F\, F'\,\mathrm{d}x = 1, \qquad E_2 - E_4 + 3E_0 = 0, \tag{13.4}$$

and diagonalizes both the generalized radial operator and the charged block

$$\mathcal{H}_{\Delta,\ell} = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \alpha^2 + \chi(1-\cos F) + \frac{\ell(\ell+1)}{x^2}. \tag{13.5}$$

The deterministic 18/18 card passes the classical radial necessary gate. It deliberately leaves lattice QC2D, the full coupled three-dimensional Hessian, quantum determinants, and global target-space unwinding open. The complete formulas are in `route_f/tex/ap_e3_charged_two_colour_soliton_proxy.tex`.

13.2 Non-extendible mixed WZW and spin refinement

Put $X = S^3 \times S^2$, choose integral generators u_3, u_2, and let $\widehat{\Omega}_n \in \widehat{H}^5(X;\mathbb{Z})$ have curvature $n\omega_3 \wedge \omega_2$ and characteristic class nu_3u_2. The bosonic action on every closed spin four-manifold, without an extension of its map, is

$$Z_{\mathrm{bos}}(M,\phi) = \exp\Big(2\pi\mathrm{i}\,\widehat{\Omega}_n(\phi_*[M])\Big). \tag{13.6}$$

Here $H^4(X;\mathbb{R}/\mathbb{Z}) = 0$, so fixed integral curvature has no ordinary flat ambiguity. Stable splitting instead gives

$$\Omega_4^{\mathrm{Spin}}(S^3 \times S^2) \simeq \mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2. \tag{13.7}$$

If ν_3 is the spin-bordism class of the inverse image of a regular point under the S^3 projection and ν_2 is the Arf class of the inverse-image spin surface under the S^2 projection, the general reduced spin refinement is

$$Z_{\epsilon_3,\epsilon_2}(M,\phi) = Z_{\mathrm{bos}}(M,\phi)(-1)^{\epsilon_3\nu_3+\epsilon_2\nu_2}. \tag{13.8}$$

The bosonic differential-character definition is complete; a microscopic APS/Dai–Freed determinant must still choose (ϵ_3, ϵ_2) before the spin-refined action is fixed.

13.3 Pure-$SU(3)$ no-go and near-miss

Adjoint breaking uses

$$\iota(A, e^{i\alpha}) = \mathrm{diag}(e^{i\alpha}A, e^{-2i\alpha}), \qquad \ker\iota = \{(1,1), (-1,-1)\}, \tag{13.9}$$

so every $[SU(2) \times U(1)]/\mathbb{Z}_2$ term descending from $SU(3)$ obeys

$$2j + q = 0 \pmod 2. \tag{13.10}$$

A colour singlet therefore has even charge, proving that the original $\mathbf{1}_{+1}$ scalar cannot occur. Since $qq(\phi^\dagger)^m$ requires $2X_q = mX_\phi$, the desired $m = 2$ would require an odd doublet charge to equal an even singlet charge. This is impossible. The closest explicit branch instead has

$$\mathbf{3} \to \mathbf{2}_{+1} \oplus \mathbf{1}_{-2}, \qquad \overline{\mathbf{3}} \to \mathbf{2}_{-1} \oplus \mathbf{1}_{+2}, \qquad M_q = m + Vy, \quad M_\ell = m - 2Vy. \tag{13.11}$$

Choosing $m = -Vy$ makes $M_q = 0$ and $M_\ell = -3Vy$, while an adjoint scalar mass split can keep $\mathbf{1}_{+2}$ light. This is an actual tree-level decoupling but its minimal baryon $qq\phi^\dagger$ is an $\mathcal{O}(1)$ $SU(2)_\phi$ flavour doublet, not the Route-E $\mathcal{O}(2)$ triplet. The full proof is in `route_f/tex/ap_e3_nonextensible_wzw_su3_uv_audit.tex`.

13.4 Physical tangent fermion and the composability gate

A half-BPS non-Abelian vortex independently yields a normalizable $\mathbb{CP}^1$ orientational modulus and a worldline fermion. On $v = 1/w$,

$$\psi^v = -w^{-2}\psi^w, \qquad D_t\psi^v = -w^{-2}D_t\psi^w, \tag{13.12}$$

so the Grassmann field is physically tangent-valued. Canonical Spin^c quantization gives

$$Q_E = \sqrt{2}\,\bar\partial_E, \qquad \mathcal{H}_{\text{ground}} = H^{0,*}(\mathbb{CP}^1, E). \tag{13.13}$$

In the declared canonical Spin^c re-quantization, the tangent Clifford generator alone corresponds to $E = \mathcal{O}$ and gives one ground state. The source-selected half-form ordering has no normalizable zero mode. A separately derived $E = \mathcal{O}(2)$ gives

$$(h^0, h^1) = (3, 0), \qquad \lambda_{n,\pm} = \pm\frac{2}{\sqrt{C}}\sqrt{n(n+3)}, \qquad \text{mult} = 2n+3, \qquad n \geq 1. \tag{13.14}$$

The intrinsic 27/27 SQM audit therefore derives a physical tangent fermion, but does not derive the canonical vacuum line and is not yet in the charged-two-colour mother model. AP–E3's signed $k = +2$ WZW character composes with it only after proving the same $B = 1$, the same $\mathbb{CP}^1$ collective-coordinate map, transgressing the degree-five character along the spatial three-cycle, and pulling the resulting differential line back with degree $+2$. The degree-one Route-E portal is postponed until that theorem or an anomaly-free product compactification is available. See `route_f/tex/ap_e4_moduli_space_sqm.tex`.

14 2026-07-16 AP–E completion frontier

This section integrates the next three requested calculations. It records the common normalization and promotion logic; the standalone notes retain the full proofs, source manifests, and negative controls.

14.1 Four-dimensional action and graded $B = 1$ Hessian

The declared Euclidean lattice theory has the schematic action

$$S_E = S_{g,W/\mathrm{imp}} + S_{\mathrm{gf}} + \bar{c}M_{\mathrm{FP}}c + S_{\chi\Delta} + \overline{\psi}D_W[U, V, z]\psi, \tag{14.1}$$

where $V \in SU(2)_c$, $z \in U(1)_g$, the meson $U \in SU(2)$ is neutral, and the complex diquark has charge two. At $(\overline{V}, \overline{z}, \overline{U}, \overline{\Delta}, \overline{\psi}) = (1, 1, U_{B=1}, 0, 0)$, write the even variables as $\Phi = (A, b, \eta, d_1, d_2)$. The complete formal quadratic structure is

$$\mathbb{H}_{\mathrm{cl}} = \begin{pmatrix} K_{\Phi\Phi} & 0 & 0 \\ 0 & 0 & D_W \\ 0 & -D_W^T & 0 \end{pmatrix}, \qquad K_{\Phi\Phi} = \begin{pmatrix} K_{AA} & K_{Ab} & K_{A\eta} & K_{Ad_1} & K_{Ad_2} \\ K_{bA} & K_{bb} & K_{b\eta} & K_{bd_1} & K_{bd_2} \\ K_{\eta A} & K_{\eta b} & K_{\eta} & K_{\eta d_1} & K_{\eta d_2} \\ K_{d_1 A} & K_{d_1 b} & K_{d_1 \eta} & K_{\Delta} & 0 \\ K_{d_2 A} & K_{d_2 b} & K_{d_2 \eta} & 0 & K_{\Delta} \end{pmatrix}, \tag{14.2}$$

together with the ghost block. The one-loop functional and its fermionic collective-coordinate curvature are

$$\Gamma^{(1)} = \frac{1}{2}\log\det{}' K_{\Phi\Phi} - \log\det{}' M_{\mathrm{FP}} - \Re\log\det D_W, \tag{14.3}$$

$$\partial_i\partial_j\Gamma_f^{(1)} = \Re\,\mathrm{Tr}\big(D^{-1}D_iD^{-1}D_j - D^{-1}D_{ij}\big)\,. \tag{14.4}$$

This is the requested full *formal* gauge–meson–ghost–fermion Hessian. The numerical certificate is deliberately smaller: it uses a 2^4 trivial-link background, substitutes a frozen positive Gauss–Newton meson block for the full Skyrme Jacobi operator, and evaluates one fermion curvature direction. It therefore cannot be called dynamical lattice QC2D or the complete nonlinear Hessian.

The 24/24 card gives

$$\begin{aligned} B(0) &= 0.999981468235, \\ \lambda_{\min}(K_{\eta,\mathrm{test}}) &= 0.621297848377, \qquad \lambda_{\min}(K_{\Delta}) = 0.694069172619, \\ \lambda_{\min}(K_{\Delta,\mathrm{control}}) &= -0.349308273811, \qquad \sigma_{\min}(D_W) = 0.702570765924, \end{aligned} \tag{14.5}$$

with gauge/ghost ξ-corrected residual 1.42×10^{-14}, Schur solve residual 6.30×10^{-16}, log-determinant curvature relative error 1.32×10^{-8}, and free Wilson/Symanzik observed orders 1.99861/3.99628. Monte Carlo, measure positivity, the exact Skyrme Jacobi zero-mode projection, continuum renormalization, FR quantization, and finite-amplitude stability all remain false gates. Full details are in `route_f/tex/ap_e5_4d_qc2d_lattice_hessian.tex`.

14.2 Two torsion generators and regulator-dependent phases

Explicit representatives of the reduced bordism generators are

$$\begin{aligned} G_3 &= (S^1_{\mathrm{R}} \times S^3, \mathrm{pr}_{S^3}, \mathrm{pt}), \\ G_2 &= (T^2_{\mathrm{RR}} \times S^2, \mathrm{pt}, \mathrm{pr}_{S^2}). \end{aligned} \tag{14.6}$$

Their transverse inverse images are the nonbounding periodic circle and the odd-spin torus. Hence

$$\mathrm{ind}_2\, D_{S^1_{\mathrm{R}}} = 1, \qquad \mathrm{ind}_2\, D^+_{T^2_{\mathrm{RR}}} = \mathrm{Arf}(T^2_{\mathrm{RR}}) = 1. \tag{14.7}$$

The chirality-paired ambient four-dimensional product spectra define only a factorized reference phase $(+1,+1)$. Stacking the two explicitly localized defect regulators instead gives the complete character table

$$Z_{\epsilon_3,\epsilon_2}(G_3) = (-1)^{\epsilon_3}, \qquad Z_{\epsilon_3,\epsilon_2}(G_2) = (-1)^{\epsilon_2}. \tag{14.8}$$

Thus all four torsion assignments are realizable with the same de Rham curvature. This proves regulator non-uniqueness; it does not select the charged-QC2D phase. That selection still requires the complete Dirac–Yukawa operator, heavy mass signs, and its odd-dimensional mapping-torus APS/Dai–Freed problem.

14.3 A semisimple replacement for the diagonal quotient

For $[SU(2)_c \times U(1)]/\mathbb{Z}_2$, every representation obeys

$$2j + q = 0 \pmod 2. \tag{14.9}$$

It admits $\mathbf{2}_{+1}$ but forbids $\mathbf{1}_{+1}$; hence any successful candidate must retain the unquotiented global form. Besides the minimal backup $SU(2)_c \times SU(2)_H$, the scan finds the simply connected simple group

$$Sp(4) \equiv USp(4) \simeq Spin(5) \tag{14.10}$$

with branching

$$\begin{aligned}
\mathbf{4} &\longrightarrow \mathbf{2}_0 \oplus \mathbf{1}_{+1} \oplus \mathbf{1}_{-1},\\
\mathbf{5} &\longrightarrow \mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0,\\
\mathbf{10} &\longrightarrow \mathbf{1}_{+2} \oplus \mathbf{2}_{+1} \oplus \mathbf{3}_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_{-2}.
\end{aligned} \tag{14.11}$$

Two Dirac $\mathbf{5}$'s provide the two light quark flavours. Two, not one, complex scalar $\mathbf{4}$'s are required so their copy index retains the global $SU(2)_\phi$ doublet. Then

$$qq(\phi_I^\dagger \phi_J^\dagger)_{\rm sym} \in \mathbf{3}_\phi \tag{14.12}$$

is the charge-neutral exactly-two $\mathcal{O}(2)$ operator at the representation-theory level. This does not rederive the earlier unit-cell Gauss rule.

One real adjoint $\mathbf{10}$ breaks the second $SU(2)$ to $U(1)$. For $M_q = m + q\,yV$, choosing $m = -yV$ keeps $q = +1$ light and makes the $q = -1, 0$ partners heavy at tree level. The displayed field content has

$$b_0^{Sp(4)} = 11 - \frac{4}{3}(2) - \frac{1}{6}(3) - \frac{1}{3}(1) = \frac{15}{2} > 0. \tag{14.13}$$

Vectorlike and Witten-parity checks pass. The tuning is not radiatively protected, however, and neither the full $\Omega_5^{\rm Spin}(BSp(4))$ character nor the heavy determinant threshold has been computed. The algebraic light coefficient $N_c X_q = +2$ is therefore not yet an all-scale orientation theorem. See `route_f/tex/ap_e3_aps_global_form_search.tex`; its card passes 58/58.

14.4 Same-soliton SQM, Callias line, CPT, and WZW descent

For a hypothetical same-soliton Fredholm family

$$\mathscr{D}_m^+ = D_{A(m)}^+ + \mathscr{Y}_m, \qquad m \in \mathcal{M}_1 \simeq \mathbb{CP}^1, \tag{14.14}$$

let the positive-mass eigenline on $S^2_\infty \times \mathbb{CP}^1$ satisfy

$$c_1(\mathcal{E}_+) = px + qy. \tag{14.15}$$

The families-Callias push-forward gives, in the fixed orientation convention,

$$\operatorname{rank}\operatorname{Ind}\mathscr{D}^+ = \epsilon p, \qquad c_1(\det\operatorname{Ind}\mathscr{D}^+) = \epsilon pq\, y. \tag{14.16}$$

Thus a rank-one $\mathcal{O}(+2)$ determinant template requires $\epsilon p = 1, q = 2$. The finite Berry/kernel-family regression obtains $c_1 = 1.999999999999981$ with unit nonzero gap. It is a topology template, not the missing four-dimensional Yukawa operator or its Callias index.

In one fixed Dolbeault polarization, the conjugate triplet is fixed by Serre duality:

$$E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee = \mathcal{O}(-2) \otimes \mathcal{O}(-2) = \mathcal{O}(-4), \qquad (h^0, h^1) = (0, 3). \tag{14.17}$$

The naive inverse $\mathcal{O}(-2)$ instead has $(h^0, h^1) = (0, 1)$. Equation (14.17) closes the mathematical fixed-polarization map, not its microscopic CPT regulator.

On framed configuration space, spatial fiber integration is intrinsically defined:

$$\widehat{\mathcal{L}}_B = \int_{\Sigma_3} \mathrm{ev}^*\, \widehat{\Omega}_5 \in \widehat{H}^2(\widetilde{\mathcal{C}}_B; \mathbb{Z}). \tag{14.18}$$

It descends through $p : \widetilde{\mathcal{C}}_B \to \mathcal{C}_B$ only when

$$\widehat{\mathcal{L}}_B \in \operatorname{im} p^*, \tag{14.19}$$

which requires an equivariant differential refinement, horizontal invariant curvature, trivial vertical and large-gauge holonomy, and trivial stabilizer characters. After that descent, its pullback to a same-soliton $\mathbb{CP}^1$ is $\mathcal{O}(+2)$ precisely when

$$\int_{\Sigma_3 \times \mathbb{CP}^1} \mathrm{ev}^*\, c(\widehat{\Omega}_5) = +2. \tag{14.20}$$

The factorized conditional map yields $nB \deg(s) = 2$, but the present theory has not identified its scalar S^2 with a normalizable soliton modulus and has not proved Eq. (14.19). The same-model card therefore passes 27/27 mathematical controls while keeping every composition flag false. The full derivation is `route_f/tex/ap_e4_same_soliton_callias_descent.tex`.

14.5 Portal authorization theorem and current decision

Let

$$\begin{aligned}
C_{\mathrm{lat}} &= \text{dynamical 4d phase plus complete projected quantum Hessian,}\\
C_{\mathrm{APS}} &= \text{selected torsion phases plus protected semisimple all-scale completion,}\\
C_{\mathrm{SQM}} &= \text{same-soliton SQM, Callias/CPT line, and gauge-basic WZW descent.}
\end{aligned} \tag{14.21}$$

The user-specified ordering is encoded exactly as

$$\text{portal authorized} \quad \Longleftrightarrow \quad C_{\mathrm{lat}} \vee C_{\mathrm{APS}} \vee C_{\mathrm{SQM}}. \tag{14.22}$$

All three booleans are currently false. Consequently no portal operator is written.

When Eq. (14.22) first becomes true, the portal theorem must still exhibit a map

$$F : \mathcal{M}_1 \longrightarrow \Sigma_E, \qquad \deg F = +1, \qquad F^*\mathcal{O}_{\Sigma_E}(2) \simeq \mathcal{O}_{\mathcal{M}_1}(2), \tag{14.23}$$

fix its orientation, show that its operator is gauge and anomaly consistent, and keep every coupling below the retained bosonic, fermionic, and Landau gaps. It must also preserve the physical CPT pairing and introduce no extra zero modes. These are recorded as future prerequisites, not assumed facts.

The integrated promotion card `route_f/code/verify_ap_e5_completion_frontier.py` passes 20/20 and sets

$$\boxed{\begin{array}{c}\texttt{any\_preportal\_route\_closed=false,}\\ \texttt{degree\_one\_portal.constructed=false,}\\ \texttt{physics\_promotion\_allowed=false}\end{array}} \tag{14.24}$$

15 AP-E6 same-background audit: exact obstructions and open gates

This section supersedes the completion-frontier status immediately above. It records the next execution round without weakening the portal ordering. The compact group denoted $Sp(4)$ in the physics convention is $USp(4)$, or $Sp(2)$ in the mathematical convention used in classifying spaces.

15.1 The exact Sp(4) gauge-bordism result

The low-degree integral cohomology and spin-bordism coefficients are

$$\begin{aligned} H^*(BSp(2);\mathbb{Z}) &= \mathbb{Z}[q_1, q_2], \qquad |q_1| = 4, \qquad |q_2| = 8, \\ (\Omega_q^{\text{Spin}})_{q=0}^5 &= (\mathbb{Z}, \mathbb{Z}_2, \mathbb{Z}_2, 0, \mathbb{Z}, 0). \end{aligned} \tag{15.1}$$

In the homological Atiyah–Hirzebruch spectral sequence

$$E^2_{p,q} = H_p(BSp(2); \Omega_q^{\text{Spin}}) \Longrightarrow \Omega^{\text{Spin}}_{p+q}(BSp(2)), \tag{15.2}$$

the only nonzero term with $p + q = 5$ is

$$E^2_{4,1} = H_4(BSp(2); \mathbb{Z}_2) = \mathbb{Z}_2. \tag{15.3}$$

The possible d_2 source and target use H_6 and H_2, both zero; the d_3 target uses H_1, and the only later possible map from this torsion group to $E^r_{0,4} \simeq \mathbb{Z}$ is zero. There is no extension ambiguity on this diagonal. Hence

$$\boxed{\Omega_5^{\text{Spin}}(BSp(2)) \simeq \mathbb{Z}_2.} \tag{15.4}$$

A geometric representative is a unit $Sp(2)$ instanton on S^4 times the nonbounding spin circle. Restriction to the instanton $SU(2)_c$ gives

$$\begin{aligned} &\mathbf{4} \downarrow \mathbf{2} \oplus \mathbf{1} \oplus \mathbf{1}, \qquad \mathbf{5} \downarrow \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{1}, \\ &\mathbf{10} \downarrow \mathbf{3} \oplus \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{1} \oplus \mathbf{1} \oplus \mathbf{1}. \end{aligned} \tag{15.5}$$

With the doublet index normalized to one, the instanton indices are $(1, 2, 6)$. The corresponding mod-two characters therefore are

$$(\chi_{\mathbf{4}}, \chi_{\mathbf{5}}, \chi_{\mathbf{10}}) = (-1, +1, +1). \tag{15.6}$$

Thus the displayed two Dirac **5**'s are trivial on the unique gauge bordism generator. This closes the gauge-bordism subgate only.

It does not select the two target-space signs. Independently,

$$\widetilde{\Omega}_4^{\mathrm{Spin}}(S^3 \times S^2) \simeq \mathbb{Z}_2\{S^1_R \times S^3\} \oplus \mathbb{Z}_2\{T^2_{\mathrm{Arf}=1} \times S^2\}. \tag{15.7}$$

Without a common microscopic mass-family lift and APS boundary problem, invertible torsion sectors can flip either character. The gauge character in Eq. (15.6) and the ordered target pair are different questions.

The Euclidean note verifies gamma-five reality, two-flavour determinant positivity away from zeroes, and the Pauli–Villars moment identities

$$c_a = (1, -3, 3, -1), \qquad \sum_a c_a a^j = 0 \quad (j = 0, 1, 2). \tag{15.8}$$

It does not specify the full interpolation $K(t)$, regulator statistics, APS domain, finite local counterterms, or regulated gauge–scalar–ghost complex. Accordingly these are partial controls, not a complete Euclidean regulator.

At the chosen light branch the infrared orientation remains

$$k_{\mathrm{IR}} = N_c X_q B = 2 \cdot 1 \cdot 1 = +2. \tag{15.9}$$

For a uniform flavour action on a complete **5**, however, the algebraic charge trace is only the conditional obstruction

$$k_{\mathrm{full}} = 2(+1) + 2(-1) + 0 = +2 - 2 = 0. \tag{15.10}$$

The local pure-gauge shifts are periodic, but no unbroken-group or gravitational APS determinant ratio has been evaluated. Therefore Eq. (15.10) is not an actual heavy-determinant matching computation and neither target eta sign is selected.

15.2 One genuinely relaxed continuum B=1 field, but an off-shell cubic Hessian

The declared classical sector is enlarged to a unit $O(4)$ meson field n, a neutral breathing field s, and a quadratic charge-two field. Its continuum energy is

$$\begin{aligned} E[n,s] = \int \mathrm{d}^3x \bigg\{ &\frac{1}{2}|\partial_i n|^2 + \frac{\kappa}{2}\sum_{i<j}\big(|\partial_i n|^2|\partial_j n|^2 - (\partial_i n\cdot\partial_j n)^2\big) \\ &+ \frac{1}{2}|\nabla s|^2 + \frac{1}{2}m_s^2(s-1)^2 + \beta^2 s^2(1-n_0)\bigg\}. \end{aligned} \tag{15.11}$$

For $n = (\cos F, \widehat{\boldsymbol{x}} \sin F)$, direct variation gives

$$\begin{aligned} (r^2 + 2\kappa \sin^2 F)F'' &= -2rF' - 2\sin F\cos F(\kappa F'^2 - 1) + \frac{2\kappa \sin^3 F \cos F}{r^2} + \beta^2 r^2 s^2 \sin F, \\ s'' + \frac{2}{r}s' &= m_s^2(s-1) + 2\beta^2 s(1-\cos F). \end{aligned} \tag{15.12}$$

With $(\beta, m_s, \kappa) = (1/2, 5/2, 1)$, a nonlinear BVP solve on $R = 8, 10, 12, 16$ converges to

$$E/(4\pi) = 6.1282155, \qquad s(0) = 0.91712896, \qquad B = +1, \tag{15.13}$$

and obeys the generalized Derrick identity. The canonical 4097-point little-endian float64 array (r, F, s) has SHA-256

`81104f59a5f3f4337739fc2a217cafc8da91581c9f1fb40b4fdba6ce0f948d59`.

This proves stationarity within the coupled hedgehog sector. It is not a Palais symmetric-criticality proof for every three-dimensional variation.

For the separately declared cubic action, the analytic constrained Hessian is

$$H_{\tan} = T^T \left[H_{\mathrm{emb}} - \bigoplus_x (n_x \cdot g_x) \mathbf{1}_4 \right] T, \qquad H_{\mathrm{proj}} = P_{\mathrm{coll}} H_{\tan} P_{\mathrm{coll}}. \tag{15.14}$$

Every $n+s$ entry is differentiated from one finite-difference energy; sparse symmetry, geodesic second differences, and the translation/isorotation projector are checked independently. The quadratic diquark and free toron/ghost operators use separately declared grids and boundaries. They are audited controls, not blocks of one aggregate matrix. Thus the $n+s$ second variation is exact while the complete declared-sector sparse-Hessian gate remains false.

It is not yet a physical stability operator. At fixed $L = 8$, the sampled continuum solution gives

N	a	B_a	$\Vert \operatorname{grad} E_a \Vert_{2,\mathrm{dens}}$
9	1.000000	0.168018	0.34766
13	0.666667	0.440012	0.43018
17	0.500000	0.625354	0.42171

The background is therefore off shell for the cubic action and its topology estimator has not converged to one. The observed negative projected curvatures are useful falsifiers of premature stability claims, but they cannot be promoted to soliton instability eigenvalues. No interacting colour links, fermion determinant, four-dimensional importance sampling, or continuum quantum limit is present. Moreover, the cubic boundary samples the finite $R = 16$ profile rather than imposing the exact vacuum; at $L = 8$ its maximum boundary deviation is about 4.93×10^{-2}. This boundary mismatch is recorded as part of the off-shell diagnostic.

15.3 The same canonical field and the physical-moduli obstruction

The Yukawa/Callias audit consumes the checksum above rather than re-solving an older profile. The declared constituent Yukawa term couples to F but not to s; this decoupling is part of the action, not a silent profile replacement:

$$\mathcal{D}_E = \gamma^\mu D_\mu + M(P_R U_F + P_L U_F^\dagger), \qquad H_F = -\mathrm{i}\alpha^i D_i + \beta M\big(\cos F + \mathrm{i}\gamma_5 \boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}} \sin F\big). \tag{15.15}$$

The true localized orbit is found before any family index is assigned. If $A \in SU(2)_V$ fixes U_F, then at a radius with $\sin F \neq 0$, $A(\boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}})A^{-1} = \boldsymbol{\tau}\cdot\widehat{\boldsymbol{x}}$ for every direction. Hence A commutes with all three Pauli matrices and

$$\mathrm{Stab}_{SU(2)_V}(U_F) = \mathbb{Z}_2, \qquad \mathcal{M}_{\mathrm{rot}} = SU(2)/\mathbb{Z}_2 \simeq SO(3), \tag{15.16}$$

not $\mathbb{CP}^1$. Quotienting combined space–isospin rotations by the diagonal hedgehog stabilizer gives the same $SO(3)$, not a second orbit. The separate scalar-vacuum $\mathbb{CP}^1$ is spatially uniform, with

$$\Vert \delta_m \phi \Vert^2_{[-L,L]^3} = 8L^3 |\delta_m \phi|^2, \tag{15.17}$$

so it is a bulk Goldstone direction rather than a normalizable soliton modulus.

At infinity Eq. (15.15) has a constant invertible mass. Its positive boundary eigenbundle on S^2_∞ is therefore trivial,

$$c_1(E_+) = 0, \qquad \mathrm{ind}_{\mathrm{Callias}} = 0. \tag{15.18}$$

This static statement does not contradict spectral flow along a four-dimensional interpolation; spectral flow equal to B does not force an endpoint zero mode.

There is also a representation-level obstruction to the former two-band template. A globally nondegenerate 2×2 Hermitian mass with one positive band on $X = S^2_\infty \times \mathbb{CP}^1$ defines $f : X \to \mathbb{CP}^1$, so its eigenline satisfies

$$c_1(E_+)^2 = f^*(u^2) = 0. \tag{15.19}$$

The desired class instead obeys

$$c_1(E_+) = x + 2y, \qquad c_1(E_+)^2 = 4xy \neq 0. \tag{15.20}$$

Thus it requires at least three trivial bands, a nontrivial ambient bundle, or a genuinely infinite-dimensional family, followed by a new uniform-gap proof. On the actual $SO(3)$, $H^2(SO(3);\mathbb{Z}) = \mathbb{Z}_2$, so only a torsion line, not a free $\mathcal{O}(2)$ class, is available.

Finally, the correctly typed universal target character is

$$\widehat{\Xi}_5 = 2\ \mathrm{pr}_U^* \,\widehat{\omega}_3 \smile \mathrm{pr}_s^* \,\widehat{c}_1(\mathsf{H}) \in \widehat{H}^5(SU(2) \times \mathbb{CP}^1;\mathbb{Z}). \tag{15.21}$$

Only after pulling it back by the evaluation map may one form

$$\int_{\Sigma_3} \mathrm{ev}^* \,\widehat{\Xi}_5 \in \widehat{H}^2(\mathcal{C}_{\mathrm{restricted}};\mathbb{Z}). \tag{15.22}$$

This establishes a typed local ansatz. It does not yet construct the full equivariant differential cocycle, prove all vertical and large-gauge holonomies, extend through Higgs zeroes and defects, or pull an $\mathcal{O}(2)$ line back to a physical same-soliton $\mathbb{CP}^1$. The finite CPT determinant identity is likewise an algebraic conjugation control, not a microscopic global regulator.

15.4 AP-E6 decision

With the stronger definitions forced by this round, let

$$\begin{aligned}
C_{Sp} &= (\text{complete regulator}) \wedge (\text{both target eta signs}) \\
&\quad \wedge (\text{full heavy APS match}) \wedge (\text{strong phase}), \\
C_{B1} &= (\text{same stationary lattice solution}) \wedge (B_a \to 1) \\
&\quad \wedge (\text{full projected super-Hessian}) \wedge (\text{4d dynamics}), \\
C_{\mathrm{Cal}} &= (\text{physical moduli}) \wedge (\text{actual gapped family}) \\
&\quad \wedge (\text{CPT determinant line}) \wedge (\text{gauge-basic descent}).
\end{aligned}$$

The exact gauge-bordism subgate, continuum BVP, sparse differentiation, and moduli/two-band no-go are genuine advances, but

$$C_{Sp} = C_{B1} = C_{\mathrm{Cal}} = \text{false}, \qquad \boxed{\text{portal not started.}} \tag{15.23}$$

The canonical standalone records are `route_f/tex/ap_e6_sp4_eta_bordism_threshold.tex`, `route_f/tex/ap_e6_relaxed_b1_multigrid_hessian.tex`, and `route_f/tex/ap_e6_same_soliton_yukawa_callias.tex`, together with their machine-readable cards and the independent AP-E6 frontier verifier.

16 AP-E7 regulator, discrete topology, and determinant-family audit

This section supersedes the AP-E6 execution status while retaining every negative boundary proved there. It executes the three requested alternatives in order. The result is deliberately asymmetric: the common quadratic regulator, a pure heavy threshold, the finite-site topology theorem, the classification of the physical $SO(3)$ collective-coordinate lines, and a minimal rank-three mass family are exact. None of them by itself closes a same-model pre-portal physics lane.

16.1 Common quadratic APS/PV complex and its exact scope

For a smooth background $\bar{X} = (\bar{A}, \bar{\Sigma}, \bar{\Phi})$, let the infinitesimal gauge action be

$$\mathcal{R}_{\bar{X}}\epsilon = \left(D_{\bar{A}}\epsilon, [\bar{\Sigma}, \epsilon], -\epsilon\bar{\Phi}\right). \tag{16.1}$$

Using background Feynman gauge, the complete bare quadratic boson/ghost operators are defined without deleting mixing terms:

$$\Delta_B = S^{(2)}_{\text{bos}}[\bar{X}] + \mathcal{R}_{\bar{X}}\mathcal{R}^\dagger_{\bar{X}}, \qquad \Delta_{\text{gh}} = \mathcal{R}^\dagger_{\bar{X}}\mathcal{R}_{\bar{X}}. \tag{16.2}$$

They act on the stated H^2 Sobolev domains after background stabilizers are projected. The fermion amplitude is built from $H = \gamma_5\mathcal{K}$. Spinor charge conjugation is retained only as a reference/scalar-mass algebraic control; no fixed-Higgs Kramers symmetry is assumed for a general adjoint mass.

The common determinant prescription uses

$$c_a = (1, -3, 3, -1), \qquad \Lambda_a^2 = a\Lambda^2, \qquad a = 0, 1, 2, 3, \tag{16.3}$$

for which

$$\sum_a c_a a^r = 0 \quad (r = 0, 1, 2), \qquad \sum_a c_a a^3 = -6, \qquad \sum_a c_a \log(\lambda + a\Lambda^2) = -\frac{2\Lambda^6}{\lambda^3} + O(\lambda^{-4}). \tag{16.4}$$

At the level of zeta determinants the regulated quadratic amplitude is

$$\begin{aligned} Z^{\text{PV}}_{\text{quad}}(\Lambda) = \tau_F \prod_{a=0}^{3} {\det}'_{\theta_B}(\Delta_B + a\Lambda^2)^{-c_a/2} \, {\det}'(\Delta_{\text{gh}} + a\Lambda^2)^{c_a} \\ \times {\det}'(H^2 + a\Lambda^2)^{N_f c_a/2}, \qquad N_f = 2. \end{aligned} \tag{16.5}$$

Here $\theta_B = \pi - \varepsilon$ is the upper lateral cut at a saddle. Equation (16.5) is a determinant prescription; it is not a claim that every half-integral power has already been realized by a local second-order Grassmann PV field. The finite part subtracts the integrated supercomplex A_4 coefficient at a fixed scale and declares positive reference masses to have zero local gauge, Hopf, and gravitational theta angles.

On $Y_5 = M_4 \times [0, 1]$, the interpolation operator and its literal APS domain are

$$\begin{aligned} \mathcal{D}_5^+ &= \partial_t + H(t), \\ \text{Dom}_{\text{APS},0}(\mathcal{D}_5^+) &= \{u \in H^1(Y_5) : P_{[0,\infty)}(H_0)u|_0 = 0, \quad P_{(-\infty,0)}(H_1)u|_1 = 0\}. \end{aligned} \tag{16.6}$$

The physical determinant-sign formula uses the fixed cut 0 and gapped endpoints:

$$\tau_F[X(t)] = \exp(\mathrm{i}\pi N_f \, \text{SF}_0\{H(t)\}). \tag{16.7}$$

An arbitrary common resolvent cut still defines a Fredholm APS problem, but its spectral flow cannot replace Eq. (16.7) without the corresponding endpoint spectral-section orientation term. This distinction is part of the regulator, not a convention that may be discarded.

The constant negative-mass heavy bundle is

$$E_h = (V_c \otimes L^{-1}) \oplus \mathbf{1}, \qquad x = c_1(L), \tag{16.8}$$

so the one-flavour index is

$$I_h = \int_{M_4} \widehat{A}(TM_4)\,\mathrm{ch}(E_h)\big|_4 = \int_{M_4} \left[-c_2(V_c) + x^2 - \frac{1}{8}p_1(TM_4)\right] \in \mathbb{Z}. \tag{16.9}$$

The rank-two charged branch contributes $-p_1/12$, and the neutral singlet contributes $-p_1/24$; hence gravity has not been omitted. For two identical Dirac flavours,

$$\boxed{\tau_h^{\mathrm{pure}} = \exp(\mathrm{i}\pi\, 2I_h) = 1} \tag{16.10}$$

on every closed spin four-manifold and every unbroken $SU(2)_c \times U(1)_H$ bundle. This closes the pure unbroken-gauge/gravity threshold, not a target-dependent mixed WZW determinant.

On the separated Higgs orbit, one can consistently declare

$$h = \widehat{\Sigma}_{\mathbf{5}}/V, \qquad P_\pm = \frac{h^2 \pm h}{2}, \qquad P_0 = 1 - h^2, \qquad (\mathrm{rank}\, P_+, \mathrm{rank}\, P_-, \mathrm{rank}\, P_0) = (2,2,1). \tag{16.11}$$

A GW-compatible target source built from these projectors has even copies on both target generators and therefore conditionally yields $(+1,+1)$ in the no-stacked-sector convention. It is not promoted to a finite overlap Dirac–Yukawa theorem until the ordinary/GW chiral projectors, five-dimensional kernel, and determinant orientation are specified in one operator.

There is a sharp reason this source does not restore the original deep-UV claim. Suppose an exact separated rank-two projector $C(\Sigma)$ were continuous and $Sp(4)$-equivariant through $\Sigma = 0$. Since the complex $\mathbf{5}$ is irreducible, Schur's lemma gives

$$C(0) = c\mathbf{1}_5, \qquad \mathrm{Tr}(HC(0)) = 0, \qquad \mathrm{Tr}(HP_+) = 2. \tag{16.12}$$

If C remains a projector, $c = 0$ or 1, so its rank is zero or five, never two. Thus the exact branch projector must lose its gap, continuity, or equivariance, or acquire new UV data. This theorem does not prohibit every non-idempotent equivariant coupling; it prohibits the exact separated rank-two selector needed by the proposed light-only family. A uniform exact flavour action on the complete $\mathbf{5}$ instead has

$$\kappa_{\mathrm{full}} = 2(+1) + 2(-1) + 0 = 0, \qquad \kappa_{\mathrm{light}} = +2, \qquad \kappa_{\mathrm{heavy}} = -2. \tag{16.13}$$

Therefore neither construction is yet an all-scale derivation of $k = +2$.

16.2 Finite-site topology and the re-relaxation result

The fixed-boundary finite-site configuration space of the declared naive $n + s$ action is

$$\mathcal{C}_N = (S^3)^{(N-2)^3} \times \mathbb{R}^{(N-2)^3}. \tag{16.14}$$

Both factors are path-connected and a finite product of path-connected spaces is path-connected. Hence no continuous integer-valued function on all of $\mathcal{C}_N$ can distinguish an exact $B = 1$ component from the vacuum. A finite-a topological sector requires an admissibility restriction, topology-preserving interpolation, or infinite dislocation barrier; it is not supplied by the naive sampled action.

For diagnostics, each cube is split into the six Freudenthal tetrahedra. For a generic target $q \in S^3$, let

$$V_T = [n_{x_0}\ n_{x_1}\ n_{x_2}\ n_{x_3}], \qquad c_T = V_T^{-1} q \quad \text{for } \det V_T \neq 0. \tag{16.15}$$

For a singular tetrahedron, $\operatorname{span} V_T$ is a proper subspace of $\mathbb{R}^4$. The generic regular target is chosen outside the finite union of these spans, so a singular cell has no preimage and contributes zero; an exceptional target is equivalently handled by an arbitrarily small regular perturbation. The numerical implementation treats $|\det V_T| < 10^{-13}$ as exceptional and separately verifies three-target agreement and a nonzero-determinant margin for every counted hit. The normalized affine tetrahedron hits q exactly when all entries of c_T are positive, and its signed contribution is

$$d_T(q) = \operatorname{sgn}(p)\operatorname{sgn}\det V_T, \qquad B_{\text{geom}} = -\sum_T d_T(q). \tag{16.16}$$

Three regular targets must agree. The admissibility margin is

$$\delta_a = \min_T \min_{\lambda_i \geq 0,\, \sum_i \lambda_i = 1} \left\| \sum_{i=0}^{3} \lambda_i n_{x_i} \right\|. \tag{16.17}$$

Only on $\delta_a > 0$ is the normalized interpolation nonsingular and the degree locally constant.

Five independent full-grid relaxations give:

N	L	a	E_{initial}	$E_\star$	B_{geom}
7	6	1.00	43.95516	9.07×10^{-16}	0
9	6	0.75	54.15076	5.33×10^{-17}	0
11	6	0.60	61.09804	8.07×10^{-16}	0
9	8	1.00	43.16126	1.77×10^{-14}	0
11	10	1.00	42.97910	2.04×10^{-15}	0

The maximum final free projected-gradient density is 1.84×10^{-8}: the optimizer has converged, but to the vacuum. Pinning only $n(0) = -1$ also gives $B_{\text{geom}} = 0$ through a degenerate tetrahedron. Admissibility-guarded searches at $\delta_{\text{floor}} = 0.10, 0.05, 0.02$ retain $B_{\text{geom}} = 1$, but stop on the respective floor with gradient densities $0.473, 0.438, 0.418$. They do not find an interior stationary point.

An explicitly artificial control fixes the central $3 \times 3 \times 3$ hedgehog n-values while re-relaxing all remaining variables independently on each grid. Its free constrained residual is below 6.2×10^{-8}, its three target degrees equal one, and its exact restricted $n + s$ Hessian has positive lowest eigenvalues $2.2643, 1.8043, 1.5002, 1.1611, 0.7563$ on the five grids. The corresponding full unanchored residuals are 0.24–0.66; therefore these are restricted controls, not soliton stability eigenvalues.

On the same sites the audit also assembles a declared quadratic diquark block, free Dirichlet gauge and ghost Laplacians, and

$$K_F = D_W^\dagger D_W \succeq 0. \tag{16.18}$$

Their positive spectra are useful implementation checks. However K_F is not the second derivative of $-\log\det D_W$, the interacting gauge–meson cross blocks are absent, and a ghost operator is not an ordinary positive bosonic Hessian block. Consequently the direct sum is not a physical BRST super-Hessian. The exact finite-site conclusion is stronger and more negative: the naive action has no exact topological component, and its tested unconstrained minimizers unwind.

16.3 The actual $SO(3)$ cover and the rank-three alternative

For the true localized orbit $SO(3) \simeq \mathbb{RP}^3$, the cellular boundary maps are $0, 2, 0$. Dualizing gives

$$H^2(SO(3); \mathbb{Z}) = \mathbb{Z}_2 = \{0, t\}, \qquad t = \beta(a), \qquad 2t = 0, \tag{16.19}$$

where $a \in H^1(SO(3); \mathbb{Z}_2)$ is the cover class. The unique nontrivial complex line is the FR line

$$\mathcal{L}_{\mathrm{FR}} = SU(2) \times_{\mathbb{Z}_2,\mathrm{sign}} \mathbb{C}, \qquad (A, z) \sim (-A, -z). \tag{16.20}$$

A generator of $\pi_1(SO(3))$ lifts from $A = \mathbf{1}$ to $A = -\mathbf{1}$, so

$$\mathrm{Hol}_\gamma(\mathcal{L}_{\mathrm{FR}}) = -1, \qquad \mathrm{Hol}_\gamma(\mathcal{L}_{\mathrm{FR}}^{\otimes 2}) = +1. \tag{16.21}$$

For a real skew-adjoint family,

$$\mathrm{Hol}_\gamma(\mathrm{Pf}) = (-1)^{\mathrm{ind}_2(\mathcal{D}_{Y_\gamma})}, \qquad \mathrm{Det} \simeq \mathrm{Pf}^{\otimes 2}. \tag{16.22}$$

Topology therefore classifies the two candidates but does not compute the actual same-Yukawa mapping-torus mod-two index. The standard conditional Skyrmion rule gives

$$(-1)^{N_c B} = (-1)^{2 \cdot 1} = +1, \tag{16.23}$$

and thus favours the trivial/bosonic sector, but it is not substituted for the missing operator computation. Inversion preserves the torsion class and admits the topological Real lift

$$J[A, z] = [A^{-1}, \bar{z}], \qquad J^2 = 1, \tag{16.24}$$

while the microscopic CPT action on spin, gauge, regulator, domain, and Pfaffian orientation remains open.

The torsion carrier cannot reproduce the old $\mathcal{O}(2)$ mechanism: $t \otimes \mathbb{R} = 0$ and a flat representative has zero curvature, whereas $c_1(\mathcal{O}(2))$ is a nonzero free class. Moreover the lowest odd FR rotor block has $j = 1/2$ and full Peter–Weyl multiplicity four, not the three holomorphic sections of $\mathcal{O}(2)$.

The independent rank-three branch does remove the AP-E6 two-band obstruction at the bundle level. On $X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w$, set

$$s = (z_0 w_0^2,\ z_1 w_1^2,\ z_0 w_1^2 + z_1 w_0^2). \tag{16.25}$$

These three sections of $\mathcal{O}(1, 2)$ have no common zero. With

$$e_+ = \frac{\bar{s}}{\|s\|}, \qquad P_+ = e_+ e_+^\dagger, \qquad \mathcal{M} = m(2P_+ - \mathbf{1}_3), \quad m > 0, \tag{16.26}$$

one has the exact global identities

$$\begin{aligned} &\mathrm{spec}\,\mathcal{M} = \{+m, -m, -m\}, && \mathcal{M}^2 = m^2 \mathbf{1}_3, && \\ &c_1(E_+) = x + 2y, && c_1(E_-) = -(x + 2y), && c_2(E_-) = 4xy. \end{aligned} \tag{16.27}$$

Thus $E_+ \oplus E_- = X \times \mathbb{C}^3$, and rank three is minimal because the $\mathbb{CP}^2$ hyperplane class can have nonzero square while the $\mathbb{CP}^1$ two-band class cannot. A scan of 11,664 points gives $\min \|s\|^2 = 0.25$, projector residual 5.64×10^{-16}, spectral residual 2.89×10^{-15}, exact gap $m = 1.7$, and discrete Berry Chern numbers $(1.000000000000003, 1.999999999999998)$.

This is a genuine topological mass-family theorem, but its second $\mathbb{CP}^1_w$ factor is still a spectator rather than a localized normalizable modulus. No charged two-colour representation has yet produced its three bands or the bidegree-$(1, 2)$ Yukawa projector, and the finite matrix gap is not a uniform Fredholm gap of the same soliton operator.

16.4 AP-E7 decision

Let C_{APS} require the common regulator, the original target-family lift, both actual target eta signs, the mixed heavy determinant, and all-scale $k = +2$. Let C_{B1} require an unanchored admissible stationary sequence, the interacting physical super-Hessian, and a dynamical continuum limit. Let C_{fam} require either the actual $SO(3)$ Pfaffian route or a physical rank-three same-soliton family, together with microscopic CPT and gauge-basic WZW descent. The current result is

$$C_{\mathrm{APS}} = C_{B1} = C_{\mathrm{fam}} = \text{false}, \qquad \boxed{\text{degree-one Route-E portal not started.}} \tag{16.28}$$

The mathematical advances are retained as reusable lemmas; they are not promotion substitutes. The canonical records are the three `route_f/tex/ap_e7_*` notes, their machine-readable cards, and the independent AP-E7 execution-frontier verifier.

17 AP-E8 topology-preserving finite-grid $B = 1$ construction

Among the three proposed continuations, the topology-preserving action was chosen because it directly changes the AP-E7 finite-site no-sector blocker. The finite GW/domain-wall target kernel and the actual $SO(3)$ Yukawa mapping-torus index remain independent open lanes.

Let $\mathcal{E}_F$ be the set of unique unordered vertex pairs that co-occur in any tetrahedron of the fixed Freudenthal split. For $-1/3 < \epsilon < 1$, $\gamma > 0$, and $t > \epsilon$, define

$$b_\epsilon(t) = -\log\frac{t-\epsilon}{1-\epsilon} + \frac{t-1}{1-\epsilon}, \qquad E_a^{\mathrm{TP}} = E_a + \gamma a \sum_{\{x,y\}\in\mathcal{E}_F} b_\epsilon(n_x \cdot n_y), \tag{17.1}$$

and set the action to $+\infty$ if any pair dot is at or below ϵ. The production values are $\epsilon = 0.01, \gamma = 1$. The subtracted linear term gives

$$b_\epsilon(1) = b'_\epsilon(1) = 0, \qquad b''_\epsilon(t) = \frac{1}{(t-\epsilon)^2}, \qquad b_\epsilon(1-u) = \frac{u^2}{2(1-\epsilon)^2} + O(u^3). \tag{17.2}$$

For any Freudenthal tetrahedron whose six pair dots exceed ϵ, every barycentric point obeys

$$\left\|\sum_{i=0}^{3}\lambda_i n_i\right\|^2 \geq \epsilon + (1-\epsilon)\sum_i \lambda_i^2 \geq \frac{1+3\epsilon}{4}. \tag{17.3}$$

At $\epsilon = 0.01$, the normalized-affine norm is therefore at least 0.507444578255. The interpolation never meets zero, so its degree is locally constant. The divergent barrier prevents every continuous finite-energy path from reaching the exceptional locus and changing B.

There is also an exact finite-dimensional existence theorem. In an energy sublevel $E_a^{\mathrm{TP}} \leq C$, nonnegativity gives

$$b_\epsilon(n_x \cdot n_y) \leq \frac{C}{\gamma a}. \tag{17.4}$$

Because b_ϵ is strictly decreasing from infinity to zero, every edge has a fixed-grid margin $n_x \cdot n_y \geq \epsilon + \delta_{a,C}$. The S^3 variables are compact and the $m_s > 0$ potential is coercive in all s_x. A minimizing sequence in any nonempty admissible component therefore has an interior limit in the same component. The limit is an unanchored minimizer and all tangent-plus-scalar first variations vanish. Sampling any smooth compact-supported continuum degree-one field makes the $B = 1$ components nonempty on all sufficiently fine grids, hence supplies a grid-indexed family of such minimizers. No compactness uniform in a is implied.

On a fixed uniformly smooth test field,

$$n(x) \cdot n(x + av) = 1 - \frac{a^2}{2}|D_v n|^2 + O(a^3), \tag{17.5}$$

so the added action has the asymptotic form

$$\gamma a \sum_e b_\epsilon(n_x \cdot n_y) = \frac{\gamma a^2}{8(1-\epsilon)^2} \int \sum_v \rho_v |D_v n|^4 \, \mathrm{d}^3 x + O(a^3). \tag{17.6}$$

This estimate is not uniform on the unknown minimizers. A compact-supported, exact-boundary-compatible degree-one regression gives a global log–log slope 2.15910, $R^2 = 0.999141$, and successive local slopes $2.3356, 2.1276, 2.0695, 2.0440$.

For tangent variations $\xi_x \perp n_x$, the exact barrier second variation on one edge, with $q = n_x \cdot n_y$ and $w = \gamma a$, is

$$\begin{aligned} \delta^2(w b_\epsilon(q)) = w\{&b''_\epsilon(q)(\xi_x \cdot n_y + n_x \cdot \xi_y)^2 \\ &+ b'_\epsilon(q)[2\xi_x \cdot \xi_y - q(|\xi_x|^2 + |\xi_y|^2)]\}. \end{aligned} \tag{17.7}$$

The full Riemann Hessian must further contain the total sphere-curvature shift $Q^T[H_{\mathrm{emb}} - \mathrm{diag}(\lambda_x I_4, 0)]Q$, with $\lambda_x = n_x \cdot G_x$. AP-E8 validates Eq. (17.7); it does not yet assemble a physical aggregate super-Hessian.

Seven independent final fields, with no core or centre pin, give

N	L	a	E_a^{TP}	direct gradient density	$\min n_x \cdot n_y$
15	6	0.428571	75.090097547	1.651×10^{-7}	0.091845
17	6	0.375000	72.444298275	3.421×10^{-7}	0.090442
19	6	0.333333	70.831336466	2.028×10^{-7}	0.083178
21	6	0.300000	69.844122881	7.704×10^{-7}	0.077786
16	6	0.400000	73.627393329	2.150×10^{-7}	0.090722
21	8	0.400000	73.346860060	1.052×10^{-7}	0.095418
26	10	0.400000	73.289478682	1.228×10^{-7}	0.095975

All three regular-value targets give $B = (1, 1, 1)$ in every case. The fixed-$a = 0.4$ energy spread over $L = 6, 8, 10$ is 0.459%, but the weighted radius changes by about 18%, so size/volume convergence is not claimed. Four nearby (γ, ϵ) controls and one deterministically perturbed initial field also return admissible stationary $B = 1$ endpoints. The total-gradient, invalid-continuation-gradient, and intrinsic-barrier- Hessian residuals are 9.38×10^{-11}, 2.81×10^{-11}, and 8.16×10^{-9}. The standalone card passes 13/13.

Thus the AP-E7 statement that no unanchored admissible finite-grid stationary representative was available is superseded:

$$\boxed{C_{B1}^{\text{finite grid}} = \text{true}, \qquad C_{B1}^{\text{physical continuum}} = \text{false}.} \tag{17.8}$$

Equicoercivity, Gamma convergence, a continuum stationary limit, barrier and triangulation independence, the physical determinant Hessian, interacting gauge/ghost blocks, four-dimensional QC2D dynamics, and the quantum continuum limit remain open. Hence the full Lane-B conjunction, portal authorization, and physics promotion remain false.

18 AP-E9: scaling regimes, Gamma-limit audit, and regulator no-go

AP-E9 resolves ordinary compactness and the barrier scaling while retaining the stronger topology and complete-action obligations. Put

$$d_a = 1 - \epsilon_a, \qquad w_a = \gamma_a a, \qquad R_a = \frac{\gamma_a a^2}{d_a^2}. \tag{18.1}$$

For one admissible edge define

$$x_e = \frac{n_x \cdot n_y - \epsilon_a}{d_a}, \qquad z_e = 1 - x_e = \frac{|n_x - n_y|^2}{2d_a}. \tag{18.2}$$

The subtracted logarithmic barrier is $\phi(x_e) = -\log x_e - 1 + x_e$, with

$$\phi(1-z) = \sum_{k\geq 2} \frac{z^k}{k}, \qquad \frac{z^2}{2} \leq \phi(1-z) \leq \frac{z^2}{2(1-z)}. \tag{18.3}$$

Consequently

$$\boxed{b_{\epsilon_a}(n_x \cdot n_y) \geq \frac{|n_x - n_y|^4}{8d_a^2}.} \tag{18.4}$$

For every fixed $C > 0$, if the total barrier is at most C, inversion by the principal Lambert branch gives the exact per-edge estimate

$$x_e \geq -W_0(-e^{-1-C/w_a}) \geq e^{-1-C/w_a}. \tag{18.5}$$

Thus uniform relative edge interiority on every positive bounded sublevel holds exactly when $\inf_a w_a > 0$; a uniform absolute margin also requires $\inf_a d_a > 0$. A single rotated vertex in a vacuum collar, with $x_e = e^{-\alpha/w_a}$, gives a bounded-action countersequence when $w_a \to 0$.

Independently, the declared base coordinate Dirichlet term has a mesh-uniform positive coefficient and gives an affine H^1 bound on every fixed physical box. Rellich compactness therefore proves strong $L^2 \times L^2$ equicoercivity for every positive γ_a and $-1/3 < \epsilon_a < 1$. This is not topological equicoercivity: strong L^2 convergence does not preserve three-dimensional degree.

For a fixed smooth sphere-valued field and a periodic shape-regular mesh,

$$\mathcal{B}_a[S_a n] = \frac{R_a}{8} \int Q_{\mathcal{T}}(\nabla n)\, \mathrm{d}x + O\left(R_a \left(a + \frac{a^2}{d_a}\right)\right). \tag{18.6}$$

At fixed d and $\gamma_a = ca^{-p}$, $w_a = ca^{1-p}$ and $R_a = ca^{2-p}/d^2$. Uniform edge interiority needs $p \geq 1$, while smooth disappearance needs $p < 2$; hence

$$\boxed{1 \leq p < 2, \qquad p = 1 \text{ is the minimal matched scaling.}} \tag{18.7}$$

If $d_a \sim d_0 a^\alpha$, this becomes $p \geq 1$ and $p + 2\alpha < 2$; an absolute margin additionally rules out $\alpha > 0$.

For the barrier alone, use the explicitly declared dual-cell piecewise-constant embedding in $L^2(\Omega;\mathbb{R}^4)$. When $R_a \to 0$, its strong-L^2 Gamma limit is zero on $L^2(\Omega;S^3)$ and $+\infty$ outside; it forgets trace and degree. If $\inf R_a > 0$, the quartic lower bound gives $W^{1,4}$ and $C^{0,\beta}$ compactness, closing degree; if $R_a \to \infty$, bounded sublevels converge to the vacuum. At finite nonzero R, the displayed $Q_{\mathcal{T}}$ is only the Cauchy–Born density on fixed smooth samples. A periodic multi-cell mesh generally requires a homogenized cell formula, which is not evaluated here. A resolved degree-one bubble of radius $r_a = \max\{R_a^{1/2}, a^{1/2}\}$ has $\mathcal{B}_a = O(R_a/r_a) \to 0$ while converging in L^2 to the vacuum. Therefore a continuum-vanishing barrier cannot by itself close degree.

At the critical scaling, the uncorrected smooth-sampling tensors are algebraically different. The uniform positive Freudenthal and checkerboard five-tetrahedron quartics are

$$Q_{6,+}(A) = \sum_i |Ae_i|^4 + \sum_{i<j} |A(e_i + e_j)|^4 + |A(e_1 + e_2 + e_3)|^4, \tag{18.8}$$

$$Q_5(A) = \sum_i |Ae_i|^4 + \frac{1}{2} \sum_{i<j} \left(|A(e_i + e_j)|^4 + |A(e_i - e_j)|^4\right). \tag{18.9}$$

For $(Ae_1, Ae_2, Ae_3) = (v, 0, 0)$ their ratio is $4/3$, whereas for $(v, v, 0)$ it is 3. No scalar change of γ equates these raw Cauchy–Born tensors; equivalence of the homogenized densities remains an open cell problem.

A complete same-action Gamma theorem is conditional: if the AP-E7 base actions are equicoercive and Gamma-converge in strong $L^2 \times L^2$, every finite-energy limit has an energy-dense sequence of fixed smooth pairs, and the nodal samples of each fixed pair recover the base energy, then a slow diagonal argument together with $a^2/d_a \to 0$ and $R_a \to 0$ imply that adding the barrier leaves that limit unchanged. The discrete forward-Skyrme-minor liminf, fixed-boundary fixed-degree density, topological-current defect control, and translation tightness for growing boxes remain unproved.

The production scan re-relaxes 23 fields over six fixed-box spacings, four fixed-spacing volumes, $p = 0, 1, 2$, both checkerboard phases, the original six-tet mesh, and centered/translated starts at $a = 0.4$ and 0.3. Every case is stationary below 2×10^{-6}, admissible, and has three-target $B = (1, 1, 1)$. Nevertheless the predeclared continuum diagnostics are

diagnostic	observed	threshold	pass
relaxed $p = 1$ barrier slope	0.2196	1.0 ± 0.4	no
largest-three-volume radius spread	8.521%	5%	no
cross-mesh energy spread, $a = 0.4$	5.442%	3%	no
cross-mesh energy spread, $a = 0.3$	5.355%	3%	no
$p = 0$ versus $p = 1$ energy spread	6.446%	3%	no

The energy-only volume spread is 0.1001%, demonstrating why energy alone is an insufficient continuum gate. The card passes 8/8 implementation and provenance checks but gives

$$\begin{gathered} C_{B1}^{\text{physical continuum}} = \text{false}, \\ \texttt{regulator\_stable\_background} = \texttt{false}, \\ \texttt{same\_action\_Hessian\_allowed} = \texttt{false}. \end{gathered} \tag{18.10}$$

The regulated determinant variation remains equally unauthorized. The finite GW/domain-wall kernel and actual $SO(3)$ Yukawa mapping-torus mod-two index remain parallel lanes; the degree-one Route-E portal remains last.

19 AP-E10: exact stencil no-go, finite-R cell formula, and translation quotient

AP-E10 gives a decisive negative answer to the proposed compensated- compactness route for the actual AP-E7 one-corner forward stencil. On a three-periodic torus let

$$\theta_z = \frac{2\pi}{3}(z_1 + 2z_2), \qquad n_a(z) = \left(\sqrt{1-\eta^2 a^2}, \eta a\cos\theta_z, \eta a \sin\theta_z, 0\right). \tag{19.1}$$

Then $n_a \to e_0$ uniformly, but the target-component $(1,2)$ entry of the spatial $(1,2)$ forward minor is identically

$$\boxed{(D_1^a n_a \wedge D_2^a n_a)_{12} = \frac{3\sqrt{3}}{2}\eta^2.} \tag{19.2}$$

The continuum minor of the limit is zero. Both the coordinate Dirichlet and the AP-E7 forward Skyrme energy remain bounded. Thus the minor weak- continuity hypothesis needed in AP-E9 is false, not merely unproved. This is consistent with div–curl theory requiring compatible differential constraints [9] and with the FEEC requirement of a discrete subcomplex and bounded cochain projection [10].

A separate two-periodic construction tracks the current defect. Put

$$u_1 = (-1)^{z_1+z_2}, \quad u_2 = (-1)^{z_2+z_3}, \quad u_3 = (-1)^{z_3+z_1}, \qquad n_a = (\sqrt{1-3\eta^2 a^2}, \eta a u_1, \eta a u_2, \eta a u_3). \tag{19.3}$$

Its image lies in an open hemisphere, so geometric degree is zero, whereas the forward algebraic current is

$$J_a = -16\eta^3\sqrt{1-3\eta^2a^2} \longrightarrow -16\eta^3. \tag{19.4}$$

After multiplication by the exact-boundary cutoff $\chi = \prod_i \sin^2(\pi x_i)$ and normalization, the weak limit is the diffuse defect $-16\eta^3\chi^3 \mathrm{d}x$. It is not a concentration measure. Indeed, if s_i are the three singular values of the forward derivative, arithmetic–geometric mean gives

$$\boxed{\int |J_a|^{4/3}\mathrm{d}x \le \frac{1}{3}\int |\wedge^2 D^a n_a|^2 \mathrm{d}x.} \tag{19.5}$$

Thus singular forward-current concentration is excluded, but a diffuse lattice alias survives and the forward current is not the normalized-affine degree current. Recent two-dimensional discrete Skyrmion compactness [11] supplies a useful architecture, not a theorem for this noncompatible three-dimensional stencil.

The AP-E9 finite-R cell problem is now solved. For periodic bonds $\mathcal{B}_Y$,

$$Q_{\mathrm{hom}}(A) = \frac{1}{|Y|}\inf_{\varphi: Y\to\mathbb{R}^m} \sum_{(p,r)\in\mathcal{B}_Y} |Ar + \varphi(p+r) - \varphi(p)|^4. \tag{19.6}$$

Every fixed-direction bond graph in both the uniform six-tet and checkerboard five-tet period-two cells is cycle-balanced. Hence the mean periodic corrector increment vanishes direction by direction; convex Jensen inequality proves that $\varphi = 0$ is the exact minimizer. Therefore the homogenized densities are precisely

$$Q_6^{\mathrm{hom}}(A) = \sum_i |Ae_i|^4 + \sum_{i<j}|A(e_i+e_j)|^4 + |A(e_1+e_2+e_3)|^4, \tag{19.7}$$

$$Q_5^{\mathrm{hom}}(A) = \sum_i |Ae_i|^4 + \frac{1}{2}\sum_{i<j}\left(|A(e_i+e_j)|^4 + |A(e_i-e_j)|^4\right). \tag{19.8}$$

The two checkerboard phases agree, but six/five ratios remain 4/3 on $(v,0,0)$ and 3 on $(v,v,0)$. Thus triangulation dependence survives homogenization; a single rescaling of R cannot remove it. Random-start cell solves reproduce the closed formulas to relative residual 9.1×10^{-15}.

Finally, AP-E10 defines the translation quotient by the density $\rho = 1 - n_0$, its barycentre, centered covariance, and barycentre-aligned radial profile. Dynamic shifts are accepted only if admissibility and all three degree targets persist, after which the same action is re-relaxed. Eleven production cases extend to $a = 0.208333$ and $L = 7.5$. Every case is stationary below 2×10^{-6}, has edge margin at least 0.1064, and gives $B = (1,1,1)$. The predeclared continuum diagnostics are

diagnostic	observed	threshold	pass
joint-tail energy spread	1.656%	3%	yes
joint-tail centered-radius spread	9.612%	5%	no
joint successive profile L^1	7.072%	5%	no
translated-start profile L^1	3.500%	3%	no
five-/six-tet energy spread	5.143%	3%	no
five-/six-tet centered-radius spread	11.378%	5%	no

The production card passes 10/10, but the exact stencil counterexample and failed mesh/shape gates force

$$\boxed{\begin{gathered}\texttt{full\_Gamma\_liminf=false},\\ \texttt{regulator\_stable\_background=false},\\ \texttt{same\_action\_Hessian\_allowed=false}.\end{gathered}} \tag{19.9}$$

The required repair is a complete tetrahedral cochain action: dn as a primal one-cochain, a graded shifted cup product satisfying a discrete Leibniz rule, a positive calibrated Hodge star, and a topological three- cochain whose evaluation agrees with normalized-affine degree. The Hessian and determinant variation remain embargoed until that replacement passes its own cell and quotient continuum gates.

20 AP-E11: compatible complete-minor cochain action

AP-E11 replaces, rather than rescales, the AP-E10 stencil. On an ordered tetrahedral complex define the Alexander–Whitney front/back cup by

$$(\alpha \smile_h \beta)[v_0 \dots v_{p+q}] = \alpha[v_0 \dots v_p]\, \beta[v_p \dots v_{p+q}]. \tag{20.1}$$

The exact differential graded identities are

$$d^2 = 0, \qquad d(\alpha \smile_h \beta) = d\alpha \smile_h \beta + (-1)^p \alpha \smile_h d\beta, \qquad (\alpha \smile_h \beta) \smile_h \gamma = \alpha \smile_h (\beta \smile_h \gamma). \tag{20.2}$$

For $a^A = dn^A$, the normalized antisymmetric cochains

$$X_h^{AB} = \frac{1}{2}(a^A \smile_h a^B - a^B \smile_h a^A), \tag{20.3}$$

$$Z_h^{ABC} = \frac{1}{3!} \sum_{\pi \in S_3} \operatorname{sgn}(\pi) a^{\pi(A)} \smile_h a^{\pi(B)} \smile_h a^{\pi(C)} \tag{20.4}$$

are exactly the de Rham cochains of the P1 affine two- and three-minors. The factor 1/2 in (20.3) is the reference-triangle area.

For a k-simplex $\sigma = [i_0 \dots i_k]$, use the Whitney form

$$\mathcal{W}_h(\sigma) = k! \sum_{r=0}^{k} (-1)^r \lambda_{i_r} d\lambda_{i_0} \wedge \cdots \widehat{d\lambda_{i_r}} \cdots \wedge d\lambda_{i_k} \tag{20.5}$$

and the consistent Hodge Gram matrix $(M_k^T)_{\sigma\tau} = \int_T \mathcal{W}_h(\sigma) \wedge \star\mathcal{W}_h(\tau)$. Every M_k^T, $k = 1, 2, 3$, is positive definite; rigid rotations preserve it; and $\mathcal{W}_h d = d\mathcal{W}_h$ reconstructs affine exterior forms exactly [12, 13, 10]. The audit gives $\lambda_{\min} = 2.485 \times 10^{-2}$, rotation residual 3.55×10^{-15}, and affine reconstruction residual 2.40×10^{-16}.

The same third-cup family supplies topology. On $T = [0123]$,

$$C_h[n](T) = \epsilon_{ABCD}(n^A \smile_h a^B \smile_h a^C \smile_h a^D)(T) = \det[n_0\ n_1\ n_2\ n_3]. \tag{20.6}$$

If every tetrahedral vertex pair obeys $n_i \cdot n_j > \epsilon > -1/3$, then for $v_T = \sum_i \lambda_i n_i$,

$$|v_T|^2 \geq \epsilon + (1 - \epsilon) \sum_i \lambda_i^2 \geq \frac{1 + 3\epsilon}{4} > 0. \tag{20.7}$$

Consequently $u_T = v_T/|v_T|$ is defined and its exact topological cochain is

$$\Omega_h(T) = \frac{s_T}{2\pi^2} C_h[n](T) \int_{\Delta^3} |v_T(\lambda)|^{-4}\, d\lambda, \qquad \sum_T \Omega_h(T) = \deg u_h. \tag{20.8}$$

This is the pullback formula for the S^3 volume form; the numerator is constant on each affine tetrahedron and equals (20.6).

The compatible action is

$$\begin{aligned} E_h = \sum_T \int_T \left\{ \frac{1}{2}|Dv_h|^2 + \frac{R}{2}| \wedge^2 Dv_h|^2 + \frac{K}{2}| \wedge^3 Dv_h|^2 + m^2(1 - v_h^0) \right\} d^3x, \\ (R, K, m^2) = (1, 0.35, 0.12). \end{aligned} \tag{20.9}$$

The positive K term is the Hodge norm of the complete family (20.4); its contraction with n and radial dressing give (20.8). It closes the finite-grid topology-collapse mode without reintroducing a mesh-direction quartic.

Every periodic P1 corrector preserves the means of all first, second, and third minors. Convex Jensen inequality on the complete minor vector proves the exact cell formula

$$\boxed{Q_{R,K}^{\text{cell}}(A) = \frac{1}{2}|A|^2 + \frac{R}{2}|M_2(A)|^2 + \frac{K}{2}|M_3(A)|^2, \qquad \varphi_{\min} = 0.} \tag{20.10}$$

It is identical on the uniform six-tet and both checkerboard five-tet meshes; the optimized residual is 2.56×10^{-14} and the minor-mean residual is 1.78×10^{-15}. This is the model-specific polyconvex mechanism [14], not a scalar rescaling of AP-E10.

The Dirichlet term gives H^1 equicoercivity and forces the P1 limit back to S^3. The L^2 Hodge bounds on the compatible second and third minors, together with Piola/null-Lagrangian identities, identify their weak limits. Equation (20.7) then identifies the normalized current $\det[v_h, D_1 v_h, D_2 v_h, D_3 v_h]/|v_h|^4$ and closes degree. Convex lower semicontinuity gives the liminf; smooth nodal interpolation gives recovery. Thus the discrete theory Gamma-converges unconditionally to the fixed-degree lower-semicontinuous relaxation. Equality with the naive unrelaxed graph- norm functional still requires a topology-sensitive smooth-density theorem [15].

The production card passes 11/11. All thirteen unanchored cases have $B = (1, 1, 1)$, projected gradient density at most 1.54×10^{-7}, and pair dot at least 0.2724. The tail reaches $(N, L, a) = (37, 8, 0.222222)$; translation and mesh comparisons use $(29, 7, 0.25)$. With the same-action positive energy density as quotient measure, the diagnostics are

diagnostic	observed	threshold	pass
joint-tail energy spread	0.437%	3%	yes
joint-tail centered-radius spread	3.489%	5%	yes
joint-tail radial CDF distance	4.067%	5%	yes
translated-start energy spread	8.33×10^{-12}	1%	yes
translated-start centered-radius spread	7.16×10^{-7}	2%	yes
five-/six-tet energy spread	0.463%	3%	yes
five-/six-tet centered-radius spread	0.752%	5%	yes
five-/six-tet radial CDF distance	3.556%	5%	yes

Therefore

$$\boxed{\begin{gathered} C_{\text{cell}} = C_{\text{quotient}} = C_{\text{relaxed regulator}} = \text{true}, \\ C_{\text{graph density}} = C_{\text{classical regulator}} = C_{\text{Hessian}} = C_{\text{det}} = \text{false}. \end{gathered}} \tag{20.11}$$

The AP-E10 stencil and mesh-universality blockers are closed. The next single mathematical gate is either fixed-degree smooth density in the complete-minor graph norm or independent regularity and isolation of the selected relaxed minimizer. The classical same-action Hessian and regulated determinant remain embargoed until one of those routes succeeds.

21 AP-E12: endpoint graph density and relaxed-minimizer audit

AP-E12 makes the AP-E11 regularity gate precise. For fixed vacuum trace, define

$$\mathcal{G}_B^2 = \{u \in W^{1,2}(\Omega; S^3) : M_2(Du), M_3(Du) \in L^2,\ \deg u = B\} \tag{21.1}$$

with strong L^2 convergence of u, Du, $M_2(Du)$, and $M_3(Du)$. Graph convergence already closes degree: if u_j, u are sphere-valued, then

$$2\pi^2 |B(u_j) - B(u)| \le \|u_j - u\|_2 \|M_3(Du_j)\|_2 + |\Omega|^{1/2} \|M_3(Du_j) - M_3(Du)\|_2. \tag{21.2}$$

Therefore a target-valued graph-dense smooth sequence is automatically in the same integer degree for all sufficiently large indices. The missing theorem is target-valued endpoint graph density, not a separate degree repair.

For the AP-E11 continuum density

$$W(F) = \frac{1}{2}\{|F|^2 + R|M_2(F)|^2 + K|M_3(F)|^2\}, \tag{21.3}$$

every minor of $F + t(a \otimes \xi)$ is affine in t. Hence

$$D^2 W(F)[a \otimes \xi, a \otimes \xi] = |a|^2 |\xi|^2 + R|DM_2(F)[a \otimes \xi]|^2 + K|DM_3(F)[a \otimes \xi]|^2, \tag{21.4}$$

so the uniform Legendre–Hadamard constant is one. More strongly, periodic null-Lagrangian means give the exact identity

$$\begin{aligned} \int \{W(F + D\varphi) - W(F)\} = \frac{1}{2} \int \{&|D\varphi|^2 + R|M_2(F + D\varphi) - M_2(F)|^2 \\ &+ K|M_3(F + D\varphi) - M_3(F)|^2\}. \end{aligned} \tag{21.5}$$

This proves model-specific strong quasiconvexity but does not construct the trace-preserving S^3 recovery sequence.

The classical Cartesian approximation result is exponent-losing: $\mathrm{Cart}^{\boldsymbol{p}}$ is strongly approximable in every $\mathcal{A}^{\boldsymbol{q}}$ with $\boldsymbol{q} < \boldsymbol{p}$, not at the endpoint [16]. Area approximation controls all minors in L^1 [17]. A scale-sharp audit also shows why the direct Malý two-hole dipole does not provide the missing counterexample here. For $u(r, \theta, z) = r^\alpha \gamma(\theta)$,

$$\int |M_2(Du)|^2 \sim \int r^{4\alpha-3}\, dr < \infty \quad \Longleftrightarrow \quad \alpha > \frac{1}{2}, \tag{21.6}$$

whereas Cauchy–Schwarz forces every core filling to pay at least $cr^{4\alpha-2}$; a non-vanishing cost requires $\alpha \le 1/2$. For $r^{1/2}\log(e/r)^{-\beta}$, minor integrability requires $\beta > 1/4$ while the filling bound tends to zero. This excludes that mechanism, not all possible endpoint concentrations.

The fallback cannot presently be closed by importing generic regularity. The formal stress and classical equation are

$$P(F) = F + R\{(\operatorname{tr} G)F - FG\} + KF \operatorname{cof} G, \qquad G = F^T F, \tag{21.7}$$

$$(I - u \otimes u)\{-\operatorname{div} P(Du) - m^2 e_0\} = 0. \tag{21.8}$$

A relaxed minimizer is not authorized to satisfy (21.8) until a local fixed-trace recovery theorem is supplied. Moreover $W(F) \lesssim 1 + |F|^6$ with $(n, p, q) = (3, 2, 6)$, while the 2024 smooth partial-regularity theorem for relaxed strongly quasiconvex integrands requires $q < \min\{np/(n-1), p+1\} = 3$ [18]. It also does not include the S^3 constraint and degree relaxation.

The production card passes 9/9. Three successively finer backgrounds have maximum one-tetrahedron energy fractions 3.966×10^{-3}, 1.849×10^{-3}, and 1.173×10^{-3}. Three amplitude-0.24 tangent perturbations re-relax to energy spread 3.38×10^{-11}, maximum radial-CDF distance 5.70×10^{-7}, and field RMS 1.31×10^{-5} after quotienting integer translations and proper target $SO(3)$. This is finite-dimensional same-basin evidence only, not continuum isolation. The authorization is

$$\boxed{\begin{gathered} C_{\text{graph degree}} = C_{\text{strong QC}} = C_{\text{dipole no-go}} = \text{true}, \\ C_{\text{endpoint density}} = \text{false (not disproved)}, \\ C_{\text{weak EL}} = C_{\text{partial regularity}} = C_{\text{classicality}} = C_{\text{isolation}} = C_{\text{Hessian}} = C_{\text{det}} = \text{false}. \end{gathered}} \tag{21.9}$$

The next theorem must be an annular target-valued endpoint replacement with equiintegrable complete minors, or a reverse-Hölder estimate giving local $L^{2+\delta}$ control of the complete-minor vector followed by a controlled S^3 projection. No same-action Riemann Hessian is assembled before one of those paths also establishes classicality and continuum isolation.

22 Reproducibility ledger

The independent arithmetic/symbolic card is

`route_f/code/verify_another_physics_route_e_bridge.py`.

Run from the repository root:

```
python3 route_f/code/verify_another_physics_route_e_bridge.py
```

It writes

`route_f/output/another_physics_route_e_bridge.json`,
`route_f/output/another_physics_route_e_bridge.md`.

The current result is 27/27 mechanical checks, including

$$\max|K_{\mathrm{tr}}^2 - I/3| = 1.11 \times 10^{-16}, \qquad |2\Delta - B| = 0, \tag{22.1}$$

zero generator-invariance and canonical factor-two residuals, exact agreement between the loaded Route-E ζ card and the declared benchmark, complex weight-two covariance error 2.80×10^{-17}, the two independent Q-ball energy estimates in Sec. 6, and oscillatory-level-set residuals below 2.7×10^{-14}. The three earlier successor cards pass 18/18, 38/38, and 27/27, respectively. The new completion-frontier cards pass 24/24, 58/58, and 27/27, while their integrated authorization gate passes 20/20. The AP-E6 follow-up cards pass 39/39, 20/20, and 25/25; their fresh-manifest, same-checksum frontier conjunction passes 15/15 while keeping every complete-lane boolean false. The AP-E7 regulator, discrete-topology, and $SO(3)$/rank-three cards pass 42/42, 16/16, and 18/18, respectively. Their independent same-checksum execution-frontier conjunction passes 16/16, with the two family alternatives represented by a disjunction but every condition within each alternative retained conjunctively. All complete-lane, portal, and promotion booleans remain false. AP-E8's topology-preserving finite-grid card passes 13/13, promotes only the finite-grid sector/stationarity subcondition, and explicitly keeps the physical-continuum Lane-B conjunction, every complete lane, portal authorization, and promotion false. AP-E9's scaling/triangulation card passes 8/8, proves fixed-box strong- L^2 equicoercivity, the barrier zero-limit, and the positive-R compactness regimes, while keeping the finite-R homogenized density, numerical continuum, same-action Hessian, and regulated-determinant gates false. AP-E10's stencil/cell/quotient card passes 10/10, explicitly disproves one-corner forward-minor compensated compactness, excludes only singular forward-current concentration, computes the finite-R zero-corrector cell densities, and proves their six-/five-tet inequivalence. Its extended stationary scan keeps the physical continuum, Hessian, and determinant gates false. AP-E11's compatible-cochain card passes 11/11, constructs exact shifted-cup Leibniz algebra and positive Whitney Hodge stars, identifies normalized-affine degree with the shared third-cup contraction, proves the complete-minor five-/ six-tet cell formula, and passes every translation/triangulation quotient threshold. It promotes the relaxed regulator background only; the classical graph-norm density, Hessian, and determinant gates remain false. AP-E12's endpoint-density/regularity card passes 9/9, proves graph-norm continuity of degree, the exact strong-quasiconvexity identity, and a scale-sharp no-go for the direct Malý dipole mechanism. The endpoint density theorem is neither proved nor disproved. The available relaxed partial-regularity exponent range does not contain the present $(2, 6)$ growth, so classicality, continuum isolation, Hessian, and determinant gates remain false. AP-E13's annular/reverse-Hölder card passes 9/9 and disproves the unconditional annular endpoint replacement by an explicit geodesic-cap Wess–Zumino obstruction. It proves a corrected replacement under linear-in-radius target oscillation, while the strong-quasiconvexity reverse-Hölder attempt remains circular at its highest cutoff-stress term. Endpoint density itself is neither proved nor disproved, and all downstream classicality, isolation, Hessian, determinant, and portal gates remain false. AP-E14's WZ/Morrey/Hopf card passes 12/12. For every fixed complete-minor graph representative it proves uniform local Wess–Zumino flux decay by L^2 absolute continuity, hence closes that statement for the selected relaxed $B = 1$ minimizer without using minimality. A localized rank-one circle-valued defect proves that WZ decay, graph energy, and fixed degree do not imply linear-scale target oscillation. The exact Hopf base/fibre split isolates a positive vertical–horizontal shear and disproves a universal exact div–curl collapse of the full cutoff remainder. Classicality, continuum isolation, Hessian, determinant, and portal gates remain false. AP-E15 freezes the AP-E11 action and ends the lattice-scan branch. Its continuum card passes 11/11. It defines the canonical nonnegative relaxation defect measure μ, proves that $\mu = 0$ is equivalent to strong complete-minor graph recovery, and proves that every compact-fibre-phase-reducible vertical defect vanishes by local strong convexity and a degree-preserving comparison. The remaining mixed Hopf-base defect is not removed. Weighted cutoff decay is closed at Lebesgue-almost every centre for each fixed map, but the shrinking-radius limit is not yet uniform along a recovery sequence. Thus neither singular nor diffuse base/mixed defect is eliminated; full $\mu = 0$, local recovery, regularity, continuum isolation, Hessian, determinant, and portal gates remain false. Parallel Dirac/Callias operator mathematics carries no promotion authority. AP-E16's continuum/algebraic card passes 12/12. It constructs the common μ-almost-everywhere tangent of all five Hopf defect components and a diagonal recovery sequence stationary, after normalization, against compact exact fibre shifts. A simultaneous ordinary-field blow-up of the three minor orders exists only for target amplitude $s \sim r$ and volumetric defect mass $\mu(B_r) \sim r^3$; nonvolumetric defects require order-specific concentration variables. A normal-ball geodesic cutoff proves the conditional purified-base Caccioppoli inequality. However, the exact homogeneous generator $F_{31} = \cos\theta_1$, $F_{42} = \cos\theta_2$ remains after fibre-phase purification, has vanishing barycentric complete-minor graph vector and weighted cutoff, but carries diffuse energy $\frac{1}{2} + \frac{R}{8}$. Thus purification plus AP-E15's two-limit weighted decay does not imply $\mu = 0$. This family is not asymptotically minimizing, so the valid next gate is to transfer normalized full local quasiminimality through a recovery-compatible boundary modification and apply strong-quasiconvex rigidity in the volumetric branch, with a separate concentration-capacity or topological alternative in nonvolumetric branches. AP-E17's continuum/algebraic card passes 12/12. The AP-E16 diagonal already has normalized full relative-homotopy local quasiminimality. AP-E17 identifies the missing operation as replacement of its oscillatory trace by the affine tangent and proves a recovery-compatible, sphere-valued modification under an explicit graph-tight annulus condition (GTA). Under GTA, the exact complete-minor strong-quasiconvexity identity becomes a generalized Young-measure variance formula; quasiminimality forces both the ordinary variance and concentration mass to vanish, so the tangent is Dirac. GTA itself is not derived from the endpoint graph bounds: a rank-two microball has bounded second-minor energy but divergent weighted gluing remainder. Independently, the sharp WZ-capacity bound $\mathcal{E}_3(B_r) \geq 3K\pi^3|\mathsf{b}(r)|^2/(2r^3)$ excludes quantized point-degree concentration and splits nonvolumetric tangents into capacity-active and WZ-neutral branches. Full $\mu = 0$ and every downstream promotion remain closed until GTA is proved for the actual minimizing diagonal. The JSON explicitly retains all cross-theory identifications as non-derived bridge axioms and sets `physics_promotion_allowed=false`.

23 AP-E13: annular endpoint replacement and reverse-Hölder audit

AP-E12 proposed a target-valued annular replacement which keeps the trace on a good sphere while controlling all three L^2 complete-minor families. AP-E13 proves that its unconditional form is false. For

$$g_\varepsilon(\omega) = (\cos\varepsilon, \sin\varepsilon\,\omega) \in \mathbb{S}^3, \qquad u_{\rho,\varepsilon}(r\omega) = g_\varepsilon(\omega) \quad \text{on } A_{\rho,2\rho}, \tag{23.1}$$

one has

$$E_1 = 8\pi\rho\sin^2\varepsilon, \qquad E_2 = \frac{2\pi}{\rho}\sin^4\varepsilon, \qquad E_3 = 0. \tag{23.2}$$

The relative Wess–Zumino volume of the trace is

$$V(\varepsilon) = 2\pi\left(\varepsilon - \frac{1}{2}\sin 2\varepsilon\right) = \frac{4\pi}{3}\varepsilon^3 + O(\varepsilon^5) \mod 2\pi^2\mathbb{Z}. \tag{23.3}$$

Gluing any trace-preserving filling $U : B_\rho \to \mathbb{S}^3$ to the canonical cap and using integer degree gives

$$\int_{B_\rho} |M_3(DU)|^2 \mathrm{d}x \geq \frac{V(\varepsilon)^2}{|B_\rho|}. \tag{23.4}$$

At $\varepsilon = \sqrt{\rho}$, the shell graph energy tends to zero but the right side of (23.4) tends to $4\pi/3$. Every sphere of the shell has the same trace, so a good-radius choice cannot remove the obstruction. After zero extension to a fixed ball, the squared third-minor densities retain positive mass on shrinking sets and are not equiintegrable. This disproves the universal annular lemma, but not endpoint graph density for every fixed map.

There is a scale-compatible repair. If a trace $g : S_\rho \to \mathbb{S}^3$ lies in a normal ball about q and $\|\log_q g\|_\infty \leq \Lambda\rho$, the geodesic cone $U(r\omega) = \exp_q((r/\rho)\log_q g(\rho\omega))$ obeys

$$\begin{aligned}
\int_{B_\rho} |DU|^2 &\leq C(\rho T_1 + \Lambda^2\rho^3), \\
\int_{B_\rho} |M_2(DU)|^2 &\leq C(\rho T_2 + \Lambda^2\rho T_1), \\
\int_{B_\rho} |M_3(DU)|^2 &\leq C\Lambda^2\rho T_2,
\end{aligned} \tag{23.5}$$

where T_k are the tangential trace-minor norms. Coarea on a good sphere turns ρT_k into the corresponding shell cost. Thus a linear Morrey oscillation bound, or a comparably strong Wess–Zumino-flux decay theorem, is the missing hypothesis.

The parallel reverse-Hölder route does not close from exact strong quasiconvexity alone. The classical stress

$$P(F) = F + a_2\{(\operatorname{tr} F^T F)F - F(F^T F)\} + a_3 F \operatorname{cof}(F^T F) \tag{23.6}$$

contains a highest cutoff product

$$\frac{|u-q|}{\rho}|M_2(Du)||M_3(Du)| \leq \eta|M_3(Du)|^2 + \frac{C_\eta}{\rho^2}|u-q|^2|M_2(Du)|^2. \tag{23.7}$$

The last term needs the same Morrey bound or prior higher integrability of M_2, so using it to prove $L^{2+\delta}$ is circular. The AP-E13 card passes 9/9: at $\rho = 10^{-8}$ the critical shell norm is 6.283×10^{-8} while the filling lower bound is 4.188790188. Accordingly the conditional replacement theorem is true; the universal replacement is false by counterexample; the model-specific reverse-Hölder theorem, endpoint density, classicality, continuum isolation, Hessian, determinant, and portal gates remain false.

24 AP-E14: fixed-map WZ decay, Morrey boundary, and Hopf div–curl test

For every fixed complete-minor graph map, including the selected relaxed $B = 1$ representative,

$$\sup_x \frac{\operatorname{dist}(\int_{B_r(x)} u^*\omega_3, 2\pi^2\mathbb{Z})^2}{r^3} \leq \frac{4\pi}{3} \sup_x \int_{B_r(x)} |M_3(Du)|^2 = o(1). \tag{24.1}$$

This is uniform absolute continuity of one fixed L^1 density and uses no minimality. It does not contradict AP-E13's scale-dependent, non-equiintegrable cap family.

WZ decay is insufficient for the required linear Morrey estimate. The circle-valued cylindrical defect

$$u(r, z) = (\cos \ell(r), \sin \ell(r), 0, 0), \qquad \ell(r) = \log \log(e/r), \tag{24.2}$$

has rank-one derivative, $M_2 = M_3 = 0$, zero WZ flux, and Dirichlet tail $2\pi/\log(e/R) \to 0$, but oscillation diameter two in every axis neighbourhood. It can be localized in a vacuum region without changing a smooth exterior $B = 1$ degree. This does not prove irregularity of the minimizer; it proves that actual relaxed minimality is the missing input.

For $u = (z, w) \in \mathbb{S}^3$, the Hopf variables

$$n = (2\Re z\bar{w}, 2\Im z\bar{w}, |z|^2 - |w|^2), \qquad a = 2u^*\theta \tag{24.3}$$

obey $da = -n^*\omega_2$ in the stated standard orientation and give

$$\begin{aligned} |Du|^2 &= (|a|^2 + |Dn|^2)/4, \\ |M_2(Du)|^2 &= (|a \wedge Dn|^2 + |da|^2)/16, \\ |M_3(Du)|^2 &= |a \wedge da|^2/64, \qquad B = (16\pi^2)^{-1} \int a \wedge da. \end{aligned} \tag{24.4}$$

Only the signed last integral has exact phase cancellation. On T^3, take

$$u = (e^{i\chi} \cos(t/2), e^{i\chi} \sin(t/2)), \qquad t = A \sin x_1, \quad \chi = B \sin x_2. \tag{24.5}$$

This lift has $da = a \wedge da = 0$ but

$$\int_{T^3} |M_2(Du)|^2 = A^2B^2\pi^3/2 > 0. \tag{24.6}$$

It therefore disproves a universal exact div–curl collapse of the full positive cutoff remainder, whose exact Hopf form is $|u - q|^2(|a \wedge Dn|^2 + |da|^2)/(16r^2)$.

The AP-E14 production card passes 12/12. Three independently re-relaxed compatible-action grids are stationary, admissible, and have degree $[1, 1, 1]$; their maximum edge-angle quotients decrease $3.767 \to 3.464 \to 3.316$. This is evidence, not a continuum Morrey theorem. The next admissible route is an inner-variation/monotonicity theory for the lower-semicontinuous relaxation with explicit defect measure, followed by elimination of the rank-one vertical–horizontal defect and an ε-regularity theorem. Hessian, determinant, and portal remain closed.

25 AP-E15: relaxation defect and fibre-phase purification

AP-E15 freezes the AP-E11 action at

$$W(F) = \frac{1}{2}|F|^2 + \frac{R}{2}|M_2(F)|^2 + \frac{K}{2}|M_3(F)|^2, \qquad (R, K, m^2) = (1, 0.35, 0.12), \tag{25.1}$$

and authorizes no further lattice scans. For a recovery sequence with weak complete-minor graph limit, define

$$\Xi(F) = (F, \sqrt{R}\, M_2(F), \sqrt{K}\, M_3(F)). \tag{25.2}$$

The canonical nonnegative relaxation defect measure is

$$\frac{1}{2}|\Xi(Du_j)|^2 \mathrm{d}x \overset{*}{\rightharpoonup} \frac{1}{2}|\Xi(Du)|^2 \mathrm{d}x + \mu. \tag{25.3}$$

Its total mass is the gap between the relaxed and naive energies, and $\mu = 0$ is equivalent to strong convergence of all three minor levels.

For the Hopf split $n = h(u)$, $a = 2u^*\theta$, a compact phase sends $a \mapsto a + 2\mathrm{d}\chi$ while preserving n, da, boundary data, and degree. The exact local phase functional has second variation

$$D^2\mathcal{F}_u(\chi)[\phi,\phi] = \int \Big(|\mathrm{d}\phi|^2 + \tfrac{R}{4}|\mathrm{d}\phi \wedge Dn|^2 + \tfrac{K}{16}|\mathrm{d}\phi \wedge da|^2 \\ + m^2\Re(e^{i\chi}z)\phi^2\Big). \tag{25.4}$$

On a three-ball it is therefore strongly convex whenever $\lambda_r = 1 - m^2r^2/\pi^2 > 0$. If u_j is a smooth fixed-degree minimizing sequence, its optimal compact-phase energy drop satisfies

$$0 \le \Delta_j(B_r) \le \mathcal{E}(u_j) - I_B \to 0, \qquad \|\mathrm{d}\chi_j\|_2^2 \le 2\Delta_j(B_r)/\lambda_r \to 0. \tag{25.5}$$

Thus every compact-fibre-phase-reducible vertical defect is zero. This does not eliminate mixed concentration produced by simultaneous oscillation of the Hopf base.

The remaining AP-E13 cutoff defect has the exact Hopf form

$$\mathfrak{D}_u(x,r;q) = \frac{1}{16r^2}\int_{B_r(x)} |u-q|^2(|a \wedge Dn|^2 + |da|^2). \tag{25.6}$$

At almost every Lebesgue point, with $q = u(x)$, Sobolev differentiability and $|M_2|^2 \in L^1$ give $\mathfrak{D}_u = O(r^3) + o(r) \to 0$. This fixed-map limit does not commute automatically with the recovery limit. A singular μ can live on the exceptional set, and a diffuse base/mixed component is not yet tied to this estimate. The next theorem must prove

$$\lim_{r\downarrow 0}\limsup_j r^{-2}\int_{B_r(x)} |u_j - q_{j,r}|^2 |M_2(Du_j)|^2 = 0 \quad \text{for } \mu\text{-a.e. } x,$$

or an equivalent sequence-uniform excess contraction. Only after that theorem yields full $\mu = 0$, local recovery, regularity, and isolation may the same-action bosonic Hessian be assembled. Parallel Dirac/Callias mathematics on a prescribed smooth Hopf base is allowed, but determinant and portal promotion remain false.

AP-E16 correction. AP-E16 proves that the displayed two-limit weighted estimate is not sufficient by itself: an exactly phase-purified homogeneous Hopf-base Young measure satisfies it while carrying nonzero diffuse graph-energy defect. The valid replacement gate is the transfer of normalized *full local quasiminimality* to tangent Young measures, followed by strong-quasiconvex rigidity in the volumetric branch and a separate concentration alternative for nonvolumetric tangents.

26 AP-E16: tangent defects, Caccioppoli control, and a base Young measure

The AP-E15 defect has the exact Hopf decomposition

$$\mu = \mu_a + \mu_n + \mu_{an} + \mu_c + \mu_{cs}, \tag{26.1}$$

corresponding to $|a|^2$, $|Dn|^2$, $|a \wedge Dn|^2$, $|da|^2$, and $|a \wedge da|^2$ with the frozen action coefficients. At μ-almost every x_0, one normalized tangent τ carries all components:

$$\frac{(T_{x_0,r_\ell})_\#\mu_\beta}{\mu(B_{r_\ell}(x_0))} \overset{*}{\rightharpoonup} \frac{\mathrm{d}\mu_\beta}{\mathrm{d}\mu}(x_0)\tau. \tag{26.2}$$

A diagonal recovery sequence can be chosen so that its normalized compact phase-relaxation drop tends to zero on every fixed rescaled ball.

If the target amplitude is $s = r^\alpha$ and the tangent mass scales as r^d, the first-, second-, and third-minor energies scale as $s^2 r$, s^4/r, and s^6/r^3. Simultaneous field normalization requires

$$2\alpha + 1 = 4\alpha - 1 = 6\alpha - 3 = d, \qquad \text{hence} \qquad (\alpha, d) = (1, 3). \tag{26.3}$$

Nonvolumetric defects therefore require order-specific tangent measures.

In one normal target ball, the geodesic cutoff $v = \exp_q((1-\eta)\log_q u)$ gives

$$\begin{aligned}\Phi_j(s) \le C[\Phi_j(t) - \Phi_j(s)] & \\ + \frac{C}{(t-s)^2} \int_{B_t \setminus B_s} & |\log_q u_j|^2 (1 + R|Du_j|^2 + K|M_2(Du_j)|^2) \\ + Cm^2 \int_{B_t} & |\log_q u_j| + \epsilon_j(B_t).\end{aligned} \tag{26.4}$$

This closes the conditional purified-base Caccioppoli inequality, but not an excess improvement.

The exact obstruction is

$$u_N = \left(\sqrt{1 - N^{-2}(\sin^2 Nx_1 + \sin^2 Nx_2)}, 0, N^{-1} \sin Nx_1, N^{-1} \sin Nx_2\right). \tag{26.5}$$

After exact fibre-phase purification, its connection is still $O(N^{-1})$, while its homogeneous base Young measure has

$$\langle F \rangle = \langle M_2(F) \rangle = 0, \qquad \left\langle \tfrac{1}{2}|F|^2 + \tfrac{R}{2}|M_2(F)|^2 \right\rangle = \tfrac{1}{2} + \tfrac{R}{8}. \tag{26.6}$$

Its weighted cutoff tends to zero. Thus phase purification plus two-limit weighted decay is not sufficient for $\mu = 0$. The sequence is not asymptotically minimizing: its normalized full local comparison deficit is positive. The next gate is therefore a recovery-compatible boundary modification which transfers normalized quasiminimality to volumetric tangent Young measures; strong quasiconvexity can then force them to be Dirac. Nonvolumetric tangents require a separate concentration-capacity or topological alternative. Hessian, determinant, and portal remain closed.

27 AP-E17: boundary transfer, Dirac rigidity, and WZ capacity

On the volumetric set

$$0 < \vartheta(x_0) = \lim_{r \downarrow 0} \frac{\mu(B_r(x_0))}{|B_r|} < \infty,$$

choose the AP-E16 diagonal so that both the global energy excess and the strong-L^2 tangent error are negligible relative to $\mu(B_r)$. The full relative-homotopy local comparison deficit then already obeys

$$\frac{\epsilon_{j_\ell}(B_{\rho r_\ell}(x_0))}{\mu(B_{r_\ell}(x_0))} \longrightarrow 0.$$

Thus the missing operation is not quasiminimality but replacement of the oscillatory boundary trace by the affine tangent $v_0(y) = F_0 y$.

AP-E17 defines the graph-tight annulus condition (GTA): on a shrinking rescaled annulus the trace lies in one hemisphere, its complete-minor energy vanishes, and

$$\int \frac{|v_\ell - v_0|^2}{\delta_\ell^2} \left(1 + R|Dv_\ell|^2 + K|M_2(Dv_\ell)|^2\right) \longrightarrow 0.$$

Under GTA, radial projection of $q+r_\ell[\eta_\ell v_\ell+(1-\eta_\ell)v_0]$ gives a sphere-valued comparison with the exact original trace, relative homotopy, and degree, and with normalized energy $|B_1|W(F_0)+o(1)$.

For the homogeneous generalized graph Young measure (ν,λ,ν^∞), null-Lagrangian moments and the AP-E12 identity give

$$\overline{W}-W(F_0)=\frac{1}{2}\int\big(|F-F_0|^2+R|M_2(F)-M_2(F_0)|^2 \\ +K|M_3(F)-M_3(F_0)|^2\big)\mathrm{d}\nu+\frac{1}{2}\lambda(B_1). \tag{27.1}$$

Transferred quasiminimality makes the left side nonpositive, so every term vanishes:

$$\nu=\delta_{F_0},\qquad \lambda=0.$$

This eliminates volumetric defect wherever GTA holds. GTA is not yet derived from the endpoint graph bounds: a rank-two microball of radius h and amplitude $h^{1/4}$ has strong-L^2 norm tending to zero, bounded second-minor energy, and weighted gluing cost of order $h^{-3/2}$.

For nonvolumetric tangents, the local Wess–Zumino number satisfies the sharp capacity bound

$$\frac{K}{2}\int_{B_r}|M_3(Du)|^2\ge\frac{3K\pi^3}{2r^3}|\mathsf{b}_u(r)|^2.$$

Hence energy of order r^d implies $|\mathsf{b}_u(r)|=O(r^{(d+3)/2})$ and excludes a nonzero quantized point degree. The normalized tangent is either WZ-capacity-active, with a positive sextic fraction but vanishing absolute charge, or WZ-neutral, in which case any surviving obstruction is analytic rather than topological. Full $\mu=0$, classicality, Hessian, determinant, and portal remain closed until GTA is proved for the actual minimizing diagonal.

A Units and normalization checks

In $\hbar=c=1$ units, $[\Phi]=[f]=[m]=[M]=1$, $[U]=4$, $[\sigma]=3$, $[R]=-1$, while Q,λ,η are dimensionless. Therefore every term in (6.2) has mass dimension four and every term in (6.16) has dimension one. For (8.5), g and ζ_{eff} are dimensionless and Λ has dimension one.

The Q-ball charge convention differs by factors of two across the literature depending on whether $\Phi=\phi e^{\mathrm{i}\omega t}$ or $\Phi=\phi e^{\mathrm{i}\omega t}/\sqrt{2}$. Equations (6.4)–(6.16) use one convention consistently; mixing conventions would change Q, f_0, and R factors and is forbidden in the verifier.

B Why a phase portal needs a second reference

Suppose $\zeta_{\mathrm{eff}}=c\Phi^2$ and define the putative invariant $I=\zeta_{\mathrm{eff}}^*\Phi^2/(|\zeta_{\mathrm{eff}}||\Phi|^2)$. Then

$$I=\frac{c^*\Phi^{*2}\Phi^2}{|c|\,|\Phi|^4}=\frac{c^*}{|c|}, \tag{B.1}$$

which is constant. The field cannot serve as its own phase reference. A nontrivial observable requires a second spurion or matrix structure, such as M_V in $M_R=M_V+\zeta_{\mathrm{eff}}K_{\mathrm{tr}}$. If that reference does not transform, it explicitly breaks the Q-ball $U(1)$; if it does transform, one must solve the coupled background and identify an invariant relative phase. This algebra is the sharp reason that phase doubling alone is not a visibility mechanism.

C Minimal falsification criteria

The bridge program should be abandoned or revised if any of the following occurs:

1. the reduced bubble moduli space is not $\mathbb{CP}^1$, or its physical bundle has Chern number other than two;
2. the zero-mode operator has extra unpaired charged modes or does not realize $H^0(T)$;
3. the exact fixed-charge soliton has a negative Hessian mode or E/Q above an allowed charge carrier;
4. every phase in $M_V + \zeta K_{\rm tr}$ is removable, leaving no invariant quantity such as (7.10);
5. the required explicit breaking makes the soliton lifetime shorter than the intended physical process;
6. a valid selection rule cannot forbid XLH and dangerous operators;
7. the embedded model fails Route-E flavor, threshold, proton-decay, or cosmology gates;
8. a gravity carrier violates the redshift relation (2.5), lensing, or fifth-force bounds.

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